

INTERNATIONAL STANDARD



Power line communication systems for power utility applications – Part 2: Analogue power line carrier terminals or APLC



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INTERNATIONAL STANDARD



Power line communication systems for power utility applications – Part 2: Analogue power line carrier terminals or APLC

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International Standard IEC 62488-2 has been prepared by IEC technical committee 57: Power systems management and associated information exchange.

This first edition of IEC 62488-2 cancels and replaces the relevant parts of IEC 60663 and IEC 60495, which will be withdrawn at a later date.

This standard is to be used in conjunction with IEC 62488-1.

The text of this International Standard is based on the following documents:

FDIS	Report on voting
57/1867/FDIS	57/1891/RVD

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all the parts in the IEC 62488 series, published under the general title *Power line communication systems for power utility applications*, can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

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INTRODUCTION

IEC 62488 series is a family of standards dealing with all aspects of power line communication systems operating over electricity power lines.

These international standards apply to power line carrier terminals and systems (PLC) used to transmit information over power networks including extra high, high and medium voltage (EHV/HV/MV) power lines. Both analogue and digital modulation as well as narrow and broadband systems will be included.

The complexity and extensive size of present-day electricity generation, transmission and distribution systems are such that it is possible to control them only by means of an associated and often equally large and complex telecommunication system having a high order of reliability.

The control of electrical networks and transmission and reception of data are through a combination of analogue and digital communication systems controlling devices and systems distributed throughout the electrical network.

The emergence of digital communication systems for controlling the devices of the electrical distribution network enables faster data transmission. The traditional analogue communication systems mainly due to legacy applications are still extensively used.

The ability to represent the various electrical parameters as an analogue signal and/or a digital signal ensures the quality and quantitative aspects of seamless communication to be maintained throughout the electrical power network.

Therefore, by using either analogue power line communication, digital power line communication or a combination of both types of systems, seamless efficient communication may be maintained throughout the power network.

In many countries, Power Line Carrier (PLC) channels represent a main part of the utility-owned telecommunication system. A circuit which would normally be routed via a PLC channel can also be routed via a channel using a different transmission medium, such as a point to point radio or open-wire circuit. Since, in many cases, automatic switching is used, the actual rerouting, although predetermined, is unpredictable.

It is important, therefore, that the input and output signals and criteria exchanged among all terminal used in the communications system are compatible. This compatibility is also beneficial in creating the ability to interchange and interconnect terminals from different sources.

This document has been prepared to enable compatibility between APLC links from different sources or between APLC links and other transmission medium to be achieved and to define the terminal performance required in APLC networks.

POWER LINE COMMUNICATION SYSTEMS FOR POWER UTILITY APPLICATIONS –

Part 2: Analogue power line carrier terminals or APLC

1 Scope

This part of IEC 62488 applies to Amplitude Modulation Single Sideband (AM-SSB) Analogue Power Line Carrier (APLC) Terminals and Systems used to transmit information over power lines (EHV/HV/MV).

In particular this document covers basically baseband signals with bandwidths of 4 kHz and 2,5 kHz, or multiples thereof, corresponding to the same high frequency bandwidth/s for single or multi-channel APLC terminals.

Figure 1 shows a schematic representation of the scope of the IEC 62488-2 standard within a complete power line communication system installation.

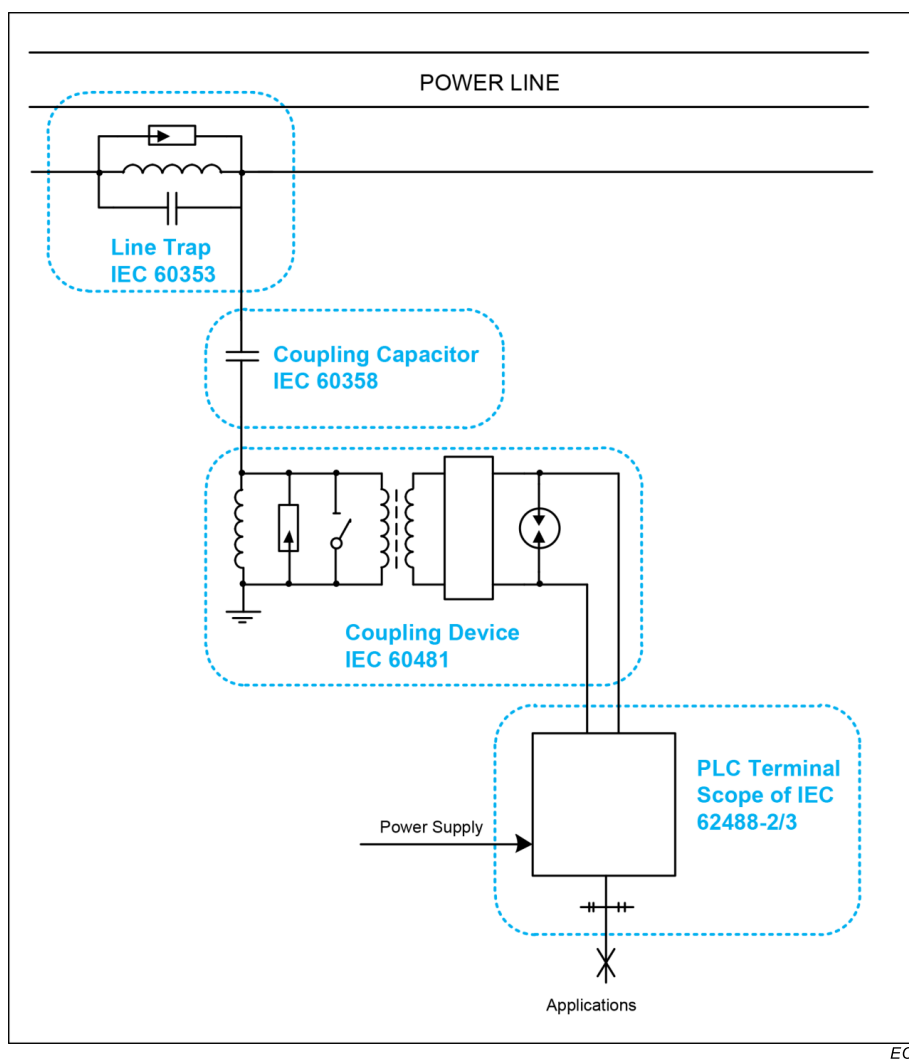


Figure 1 – Schematic representation of the scope of IEC 62488-2

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60038, *IEC standard voltages*

IEC 60068-2-1, *Environmental testing – Part 2-1: Tests – Test A: Cold*

IEC 60068-2-2, *Environmental testing – Part 2-2: Tests – Test B: Dry heat*

IEC 60068-2-6, *Environmental testing – Part 2-6: Tests – Test Fc: Vibration (sinusoidal)*

IEC 60068-2-27, *Environmental testing – Part 2-27: Tests – Test Ea and guidance: Shock*

IEC 60068-2-30, *Environmental testing – Part 2-30: Tests – Test Db: Damp heat, cyclic (12 h + 12 h cycle)*

IEC 60068-2-31, *Environmental testing – Part 2-31: Tests – Test Ec: Rough handling shocks, primarily for equipment-type specimens*

IEC 60255-27:2013, *Measuring relays and protection equipment – Part 27: Product safety requirements*

IEC 60529, *Degrees of protection provided by enclosures (IP Code)*

IEC 60721-3-1:1997, *Classification of environmental conditions – Part 3 Classification of groups of environmental parameters and their severities – Section 1: Storage*

IEC 60721-3-2:1997, *Classification of environmental conditions – Part 3: Classification of groups of environmental parameters and their severities – Section 2: Transportation*

IEC 60721-3-3:1994, *Classification of environmental conditions – Part 3: Classification of groups of environmental parameters and their severities – Section 3: Stationary use at weatherprotected locations*

IEC 60721-3-3:1994/AMD1:1995

IEC 60721-3-3:1994/AMD2:1996

IEC 60834-1, *Teleprotection equipment of power systems – Performance and testing – Part 1: Command systems*

IEC 60950-1, *Information technology equipment – Safety – Part 1: General requirements*

IEC 61000-4-2, *Electromagnetic compatibility (EMC) – Part 4-2: Testing and measurement techniques – Electrostatic discharge immunity test*

IEC 61000-4-3, *Electromagnetic compatibility (EMC) – Part 4-3 : Testing and measurement techniques – Radiated, radio-frequency, electromagnetic field immunity test*

IEC 61000-4-4, *Electromagnetic compatibility (EMC) – Part 4-4: Testing and measurement techniques – Electrical fast transient/burst immunity test*

IEC 61000-4-5, *Electromagnetic compatibility (EMC) – Part 4-5: Testing and measurement techniques – Surge immunity test*

IEC 61000-4-6, *Electromagnetic compatibility (EMC) – Part 4-6: Testing and measurement techniques – Immunity to conducted disturbances, induced by radio-frequency fields*

IEC 61000-4-8, *Electromagnetic compatibility (EMC) – Part 4-8: Testing and measurement techniques – Power frequency magnetic field immunity test*

IEC 61000-4-11, *Electromagnetic compatibility (EMC) – Part 4-11: Testing and measurement techniques – Voltage dips, short interruptions and voltage variations immunity tests*

IEC 61000-4-16, *Electromagnetic compatibility (EMC) – Part 4-16: Testing and measurement techniques – Test for immunity to conducted, common mode disturbances in the frequency range 0 Hz to 150 kHz*

IEC 61000-4-17, *Electromagnetic compatibility (EMC) – Part 4-17: Testing and measurement techniques – Ripple on d.c. input power port immunity test*

IEC 61000-4-18, *Electromagnetic compatibility (EMC) – Part 4-18: Testing and measurement techniques – Damped oscillatory wave immunity test*

IEC 61000-4-20:2010, *Electromagnetic compatibility (EMC) – Part 4-20: Testing and measurement techniques – Emission and immunity testing in transverse electromagnetic (TEM) waveguides*

IEC 61000-4-29, *Electromagnetic compatibility (EMC) – Part 4-29: Testing and measurement techniques – Voltage dips, short interruptions and voltage variations on d.c. input power port immunity tests*

IEC 61000-6-2, *Electromagnetic compatibility (EMC) – Part 6-2: Generic standards – Immunity standard for industrial environments*

IEC 61000-6-4:2006, *Electromagnetic compatibility (EMC) – Part 6-4: Generic standards – Emission standard for industrial environments*

IEC 61000-6-4:2006/AMD1:2010

IEC 61000-6-5:2015, *Electromagnetic compatibility (EMC) – Part 6-5: Generic standards – Immunity for equipment used in power station and substation environment*

IEC 62488-1:2012, *Power line communication systems for power utility applications – Part 1: Planning of analogue and digital power line carrier systems operating over EHV/HV/MV electricity grids*

CISPR 16-1-1:2015, *Specification for radio disturbance and immunity measuring apparatus and methods – Part 1-1: Radio disturbance and immunity measuring apparatus – Measuring apparatus*

CISPR 16-1-2:2014, *Specification for radio disturbance and immunity measuring apparatus and methods – Part 1-2: Radio disturbance and immunity measuring apparatus – Coupling devices for conducted disturbance measurements*

CISPR 16-1-4:2010, *Specification for radio disturbance and immunity measuring apparatus and methods – Part 1-4: Radio disturbance and immunity measuring apparatus – Antennas and test sites for radiated disturbance measurements*

CISPR 16-2-1:2014, *Specification for radio disturbance and immunity measuring apparatus and methods – Part 2-1: Methods of measurement of disturbances and immunity – Conducted disturbance measurements*

CISPR 16-2-3:2016, *Specification for radio disturbance and immunity measuring apparatus and methods – Part 2-3: Methods of measurement of disturbances and immunity – Radiated disturbance measurements*

CISPR 14-1:2016, *Electromagnetic compatibility – Requirements for household appliances, electric tools and similar apparatus – Part 1: Emission*

CISPR 22:2008, *Information technology equipment – Radio disturbance characteristics – Limits and methods of measurement*

3 Terms, definitions and abbreviations

3.1 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 62488-1 and the following apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

3.1.1

absolute power level

power of a signal expressed in decibels referred to 1 mW as defined by Equation (1)

$$X[\text{dBm}] = 10 \times \log_{10}(P/1 \text{ mW}), \quad (1)$$

where

P = signal power

3.1.2

APLC equipment

PLC equipment with an LF interface for each LF channel, applying AM-SSB modulation to the signals at the input of each LF interface and transmitting over the power line the modulated signals, which occupy adjacent frequency bands fully covering the HF transmission band of the PLC

Note 1 to entry: The LF channel bandwidths are typically either 4,0 or 2,5 kHz, but can be different.

Note 2 to entry: In addition to the mentioned interfaces other interfaces may be present such as data interfaces, LF teleprotection and control interfaces or teleprotection command interfaces.

3.1.3

basic high frequency band

elementary subdivision of the high frequency range or part thereof allocated to a single APPL transmit or receive high frequency channel

3.1.4

effective transmitted voice band

that part of the voice-frequency baseband used for telephone communication, not including the telephone signalling channel

3.1.5

level reference point

point in a system with the relative level 0 dBr

Note 1 to entry: More specifically, a sinusoidal reference signal of 1020 Hz in the voice frequency band is thought to pass the signal path under consideration with such amplitude that its absolute level is 0 dBm at the 0 dBr point. [ITU-T G.100.1]

Note 2 to entry: The level reference point may exist physically or only hypothetically.

3.1.6

LF channel

frequency band used for the signal transmission of a low frequency interface

3.1.7

load capacity, <for the multi-channel terminal>

maximum single frequency sine wave power expressed in dBm0 which a multi-channel amplifier must transmit without appreciable overloading

3.1.8

mean high frequency band output power

output power of an APLC terminal at nominal high frequency band terminal averaged over a time sufficiently long compared with the cycle time of the lowest modulation frequency and during which this average power assumes the highest value for which the terminal has been designed

3.1.9

nominal high frequency band

frequency band in which a particular APLC transmitter or receiver is operating within the carrier-frequency range

Note 1 to entry: The nominal high frequency band is also referred to as nominal carrier frequency band.

3.1.10

nominal high frequency band output power

output power of an APLC terminal at nominal high frequency band expressed as the peak envelope power (PEP) for which the terminal has been designed, compatible with the requirements for spurious emissions and inter-channel interference (for multi-channel PLC terminals), available at the high frequency interface output across a resistive load equal to the nominal impedance

3.1.11

power ratio

ratio between the powers P_1 and P_2 of two signals

Note 1 to entry: The power ratio X_{12} is usually expressed in decibels (dB) and is calculated according to equation (2)

$$X_{12}[\text{dB}] = 10 \times \log_{10}(P_1/P_2). \quad (2)$$

3.1.12

relative power level

power level at a point on a circuit given by the expression $10 \times \log_{10}(P/P_0)$ dBr, where P represents the power of a sinusoidal signal at the reference frequency 1020 Hz at the point concerned and P_0 the power of that signal at the 0 dBr reference point [ITU-T P.10/G.100 R-8, ITU-T G.100.1]

Note 1 to entry: The relation between the relative power level P_{rel} , the absolute power level P and the system power level P_0 is given by Equation (3):

$$P_{\text{rel}}[\text{dBr}] = P[\text{dBm}] - P_0[\text{dBm0}] \quad (3)$$

Note 2 to entry: The relative power level depends on the measuring point and not on the type of signal.

3.1.13**spurious emissions**

emissions on a frequency, or on frequencies, which are outside the necessary bandwidth and the level of which may be reduced without affecting the corresponding transmission of information

Note 1 to entry: Spurious emissions include harmonic emissions, parasitic emissions, intermodulation products and frequency conversion products but exclude out-of-band emissions [ITU-R SM.329-12].

3.1.14**system power level**

power level expressed in dBm₀, such that x dBm₀ means x dB above or below 1 mW at the level reference point of the system, according to whether x is positive or negative, i.e. x dB above or below the level reference signal of 1020 Hz at any point of the system

Note 1 to entry: Networks are often designed to carry different types of signals (speech, modem, fax, etc.) at different levels, expressed in dBm₀ [ITU-T G.100.1].

3.1.15**system psophometric power level**

psophometrically weighted level expressed in dBm_{0p} as defined according to ITU-T O.41

Note 1 to entry: The term dBm_{0p} is used in the same way as dBm₀ when the level is measured with a psophometer instead of a flat-response level measuring set. The psophometric weighting is used only in connection with the specification of noise levels on voice channels. The equivalent noise bandwidth of the psophometrically-weighted filter is about 1,8 kHz.

3.2 Abbreviations

AC	Alternating Current
AM-SSB	Amplitude Modulation SSB
APLC	Analogue PLC
DC	Direct Current
DPLC	Digital PLC
DSP	Digital Signal Processing, Digital Signal Processor
E&M	Ear & Mouth
FAR	Fully Anechoic Room
FDM	Frequency Division Multiplexing, Frequency Division Multiplexer
FSK	Frequency Shift Keying
FXO	Foreign Exchange Office
FXS	Foreign Exchange Subscriber
HF	High Frequency
LAN	Local Area Network
LCL	Longitudinal Conversion Loss
LF	Low Frequency
MIB	Management Information Base
OATS	Open Area Test Site
OSB	Output Signal Balance
PBX	Private Branch Exchange
PEP	Peak Envelope Power
PLC	Power Line Carrier
RMS	Root Mean Square
Rx	Receiver, receive

SAC	Semi Anechoic Chamber
SNMP	Simple Network Management Protocol
SSB	Single Side Band
Tx	Transmitter, transmit
VF	Voice Frequency
WAN	Wide Area Network
2W	Two Wire
4W	Four Wire

4 Low frequency side interfaces

4.1 General

All interfaces described in Clause 4 are optional. However, as stated in the definition of 3.1.2, it shall be possible to configure APLC equipment such that it has one low frequency interface for each LF channel.

Figure 2 shows the generic architecture of an APLC terminal with the LF interfaces on the left and the RF interface on the right. The type of interface – digital (⏏) or analogue (⏚) – is indicated. In addition, two internal conceptual signals appear in this figure: (1) The “Baseband signal”, representing the FDM multiplexed signal of all LF signals and (2) the “Passband signal”, representing the baseband signal converted to the RF band by SSB modulation. These two signals are generally not accessible by the user. In case only one LF interface is present, the function of the block “Frequency Division Multiplexer / Demultiplexer” degenerates to a simple through connection.

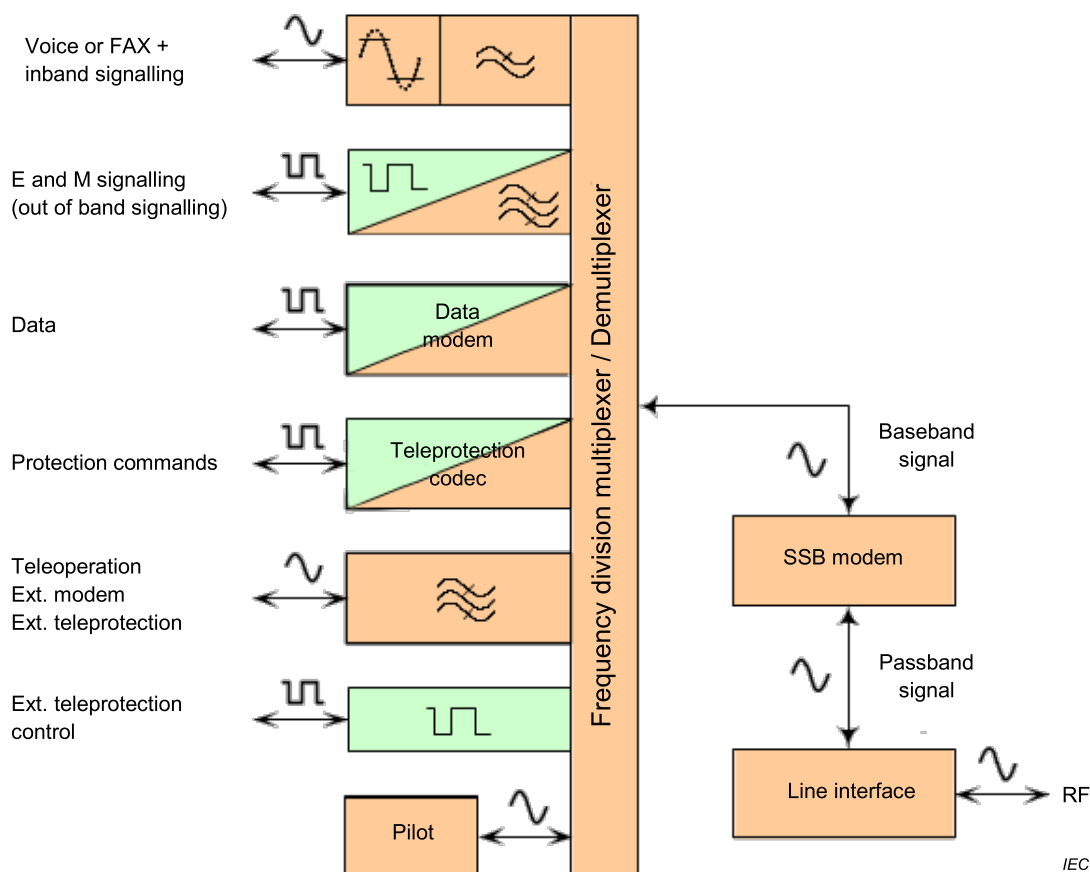


Figure 2 – Generic architecture of an APLC terminal

4.2 Analogue interfaces

4.2.1 General

In 4.2, a description of the analogue interfaces is given. For each of them the key parameters and the recommended characteristic values are indicated. All levels are specified as absolute power levels (dBm) unless otherwise noted.

4.2.2 Voice frequency band

The effective transmitted voice-frequency band is 300 to 3 400 Hz according to ITU-T G.232. Other recommended values for the effective transmitted voice band are:

- 300 to 2 400 Hz,
- 300 to 2 000 Hz.

4.2.3 Nominal impedance

The nominal impedance of all analogue low-frequency inputs and outputs shall be 600 Ω balanced.

4.2.4 Return loss

The return loss for all voice-frequency interfaces shall not be less than 14 dB within the effective transmitted frequency band.

4.2.5 Degree of unbalance to Earth

For all voice-frequency inputs longitudinal conversion loss and outputs signal balance shall not be less than 40 dB within the effective transmitted frequency band.

4.2.6 ITU-T voice channel interface

4.2.6.1 Description

One of the most used interfaces is related to the voice applications. Typically, it is used for interconnection to/from the multiplexor transit of subscriber service. One or more voice channel interfaces could be present in each terminal if needed.

4.2.6.2 Voice signal levels

- The minimum range of the transmit input level shall be –17 to 0 dBm
- The minimum range of the received output level shall be –7 to +4 dBm

4.2.7 Subscriber interface

4.2.7.1 Description

In some cases it is requested to make a subscriber connection to a PBX from a remote location. To make this possible APLC links provide for the telephone connection between PBX and two-wire telephone set by configuring a PBX interface at the local APLC terminal and a subscriber interface at the remote APLC terminal.

In Figure 3 a possible application scheme is given.

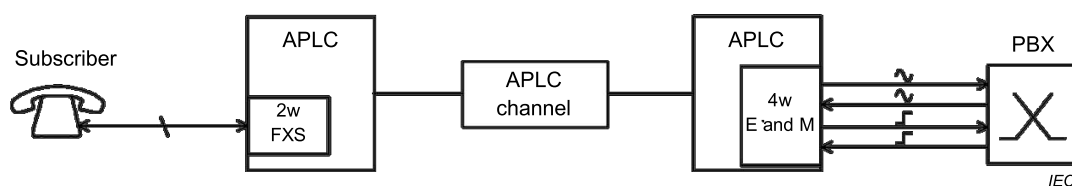


Figure 3 – Subscriber PBX interfaces local and remote

There is also the possibility to have a 2W interface to the PBX (2W-FXO interface), as an alternative to the 4W interface shown in the right side of Figure 3.

4.2.7.2 Voice signal levels

The relative two-wire/four-wire levels used for voice differ from country to country. They shall preferably be within the following recommended ranges:

- The minimum range of the transmit input level shall be –17 to 0 dBm
- The minimum range of the received output level shall be –7 to +4 dBm

4.2.8 PBX interconnection interface

4.2.8.1 Description

When the interconnection between two PBXs is needed, APLC is configured to provide the four-wire interface including the correct signaling.

In Figure 4 a possible application scheme is given.

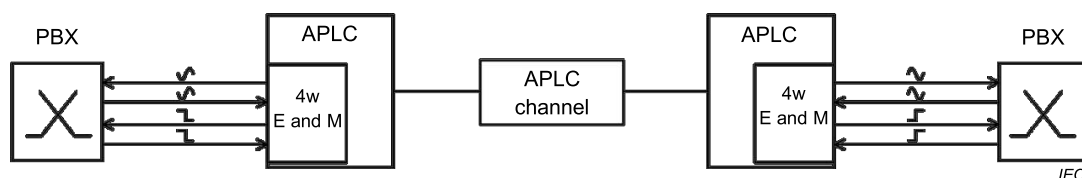


Figure 4 – Interfaces for PBX trunk interconnection through APLC link

4.2.8.2 Voice signal levels

The relative four-wire levels used for voice differ from country to country. They shall preferably be within the following recommended ranges:

- The minimum range of the transmit input level shall be –17 to 0 dBm
- The minimum range of the received output level shall be –7 to +4 dBm

4.2.9 Narrowband telegraphic channel interface

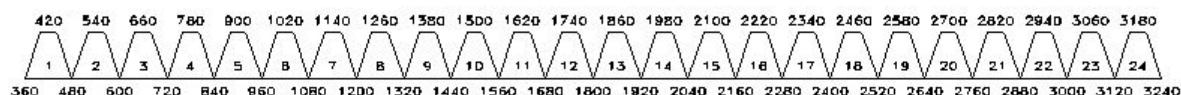
4.2.9.1 Description

Subclause 4.2.9 refers to the transit of low symbol rate modulated telegraphic channels (i.e. 50, 100, 200, 600, 1 200 Bauds) typically used for carrying remote control and metering signals from the RTUs to the SCADA central systems.

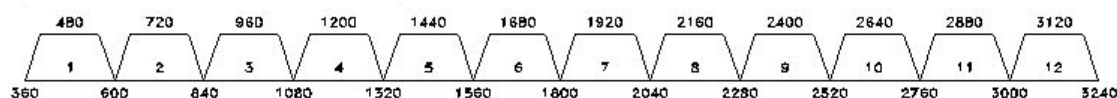
4.2.9.2 Narrowband telegraphic channelling

For asynchronous data transmission within ITU-T voice channels and according to the type of modulation used, a frequency division multiplexing and channelling plan as shown in Figure 5 is most commonly used.

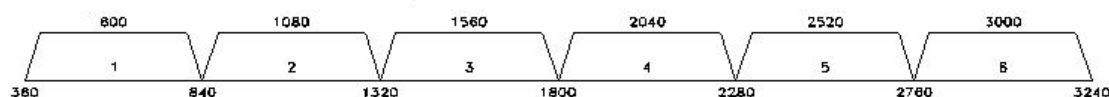
50 Bd/120 Hz CCITT R36 Channels



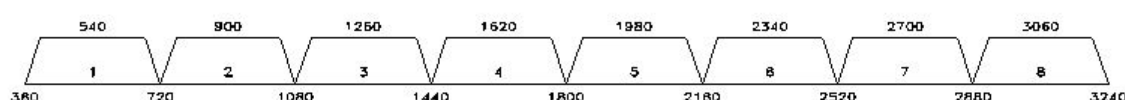
100 Bd/240 Hz CCITT R37 Channels



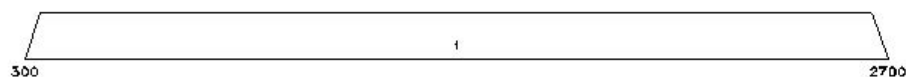
200 Bd/480 Hz CCITT R38A Channels



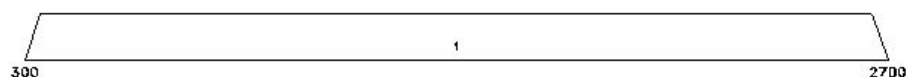
200 Bd/360 Hz CCITT R38B Channels



600 Bd CCITT V23 Channels



1 200 Bd CCITT V23 Channels



IEC

Figure 5 – Low symbol rate ITU-T telegraphic channelling

The bandwidth used within the voice channel ranges for each telegraphic channel from 120 Hz up to 2 400 Hz.

This allows the multiplexing of one or more FSK modulated signals with symbol rates from 50 Baud to 1200 Baud.

4.2.9.3 Telegraphy signal levels

The relative four-wire levels used for telegraphic modulated signals shall preferably be within the following recommended ranges:

- The minimum range of the transmit input level shall be –20 dBm to 0 dBm
- The minimum range of the received output level shall be –20 dBm to 0 dBm

4.3 Analogue teleprotection system interface

4.3.1 Description

A typical goal of a selective analogue teleprotection device consists of accelerating, delaying, locking or conditioning the protection relay (line distance protection, insulation gas low pressure protection) installed in the field at both sides of an APLC link.

This is achieved by transmitting a command from one teleprotection device to the other by a suitable transmission system, fast, safe and reliable in order to control the way the protection relay at the other end of the line section behaves.

APLC terminals optionally may have on board an integrated analogue selective teleprotection device with its interface enabling the communication with protection device.

Another possible option includes a simple analogue interface enabling the communication between an external teleprotection device with an APLC terminal.

In the first case the interface process signals/commands directly exchanged with the field protection relay. It means that no low frequency signals exchanged but simply logic criteria/commands.

In the second case the interface cooperates with an external selective teleprotection device. Therefore the interface may consist of voice-frequency input/output and control circuits.

The inputs and outputs shall preferably be decoupled from the other signal/command inputs/outputs.

4.3.2 Integrated teleprotection

In this case the functional requirements of teleprotection command systems as defined in IEC 60834-1 shall be applied.

4.3.3 Teleprotection interface frequency band

The nominal frequency band used for teleprotection service shall be specified within the frequency range of the nominal voice channel.

It can be a fully dedicated and pre-allocated frequency band or an alternative one which is allocated when the terminals switch to the teleprotection mode.

For this second operational mode – in order to reduce the risk of interference – the inhibition of all other signals present at the low frequency interface is requested.

4.3.4 Teleprotection interface impedance

- The input/output impedance shall be 600 Ω .

4.3.5 Teleprotection interface reflection

- The return loss shall be not less than 14 dB.

4.3.6 Teleprotection interface signal levels

- The minimum range of the transmit input level shall be –20 dBm to 0 dBm.
- The minimum range of the received output level shall be –20 dBm to 0 dBm.

4.3.7 Teleprotection interface control circuits

Control inputs may be provided to accomplish boosting of the teleprotection signal up to a maximum specified high frequency PEP output power and to interrupt the other signals for a short time of typically less than 500 ms during teleprotection transmission or reception.

APLC terminals shall include control circuits to shift the terminal from communication mode to teleprotection mode if a temporary band is allocated for transmitting the teleprotection signal.

The control inputs shall operate with a loop resistance of up to 500 Ω .

Control outputs are often provided to give an alarm to the external teleprotection terminal in case of failure or degradation of the APLC link.

4.4 Digital interfaces

4.4.1 Telephony signaling interface

4.4.1.1 Description

It enables interfacing with a telephone or telephone exchange. To satisfy the different telephony applications the following typical configurations shall be possible:

- automatic line extension at telephone side (FXS);
- automatic line extension at exchange side (FXO).

In some specific cases the following configurations are also requested:

- central battery;
- local battery.

In particular, it can perform the following functions:

- conversion of engaged, dialing or call criteria into transmission and return criteria for the signaler;
- generation of supply current for the telephone;
- generation / detection of an alternating ring voltage of 20/25 Hz or 50/60 Hz.

4.4.1.2 E&M signaling mechanism

With reference to Figures 3 and 4, the signaling channel shall be operated by a potential free open or closed contact or by earthing at the transmit side and shall provide a potential free contact or earth at the receive side.

The input shall operate with a loop resistance up to 500 Ω . The output contact shall be able to switch up to 72 V and 50 mA on a resistive load.

4.4.1.3 Signaling modulation and frequency band

Often signaling is transmitted using FSK modulation. To this aim a specific small portion of the frequency band is allocated within or beside the voice channel (e.g. 2 490 – 2 550 Hz). The FSK frequency of the FSK signal is shifted (for instance by ± 30 Hz) in the presence of signaling.

An alarm is often provided if the level of the signaling frequency being received is less than a predefined adjustable threshold (e.g. 3 dB to 20 dB) with respect to the nominal value.

4.4.1.4 Pulse distortion

The pulse distortion shall not exceed 10 % when the signaling channel is operated at a speed of 30 Baud.

4.4.2 Internal data modem

4.4.2.1 Description

For the data transmission, one or more serial modems may be integrated into the APLC terminal.

Each modem interface according to the technology implemented shall be compliant with the respective communication standard.

ITU-T recommendation V.24 (EIA RS-232) and recommendation V.11 standards define serial interfaces to be used for communication between digital devices.

For FSK transmission the bandwidth to be used is defined by the respective standard. It uses normally one or more parts of the typical telephone channel as shown in the previous Figure 5.

The bandwidth to be used is defined by the respective standard. The modems use the full bandwidth of an LF channel or one or more parts thereof. For FSK, the channelling is shown in Figure 5.

Other kind of modulation such as DPSK, QPSK, QAM, etc. can also be used for data transmission.

4.4.2.2 Physical layer

The serial data interfaces shall be compliant with ITU-T recommendations V.24/V.28 (EIA RS-232) or ITU-T recommendations V.11/X.24 (EIA RS-422). Commonly used connectors are shown in Figure 6 for EIA RS-232 and Figure 7 for EIA RS-422.

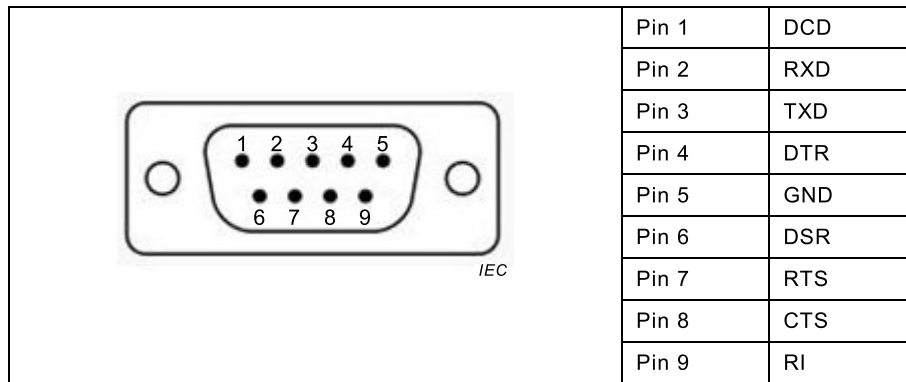


Figure 6 – Commonly used EIA RS-232 connector

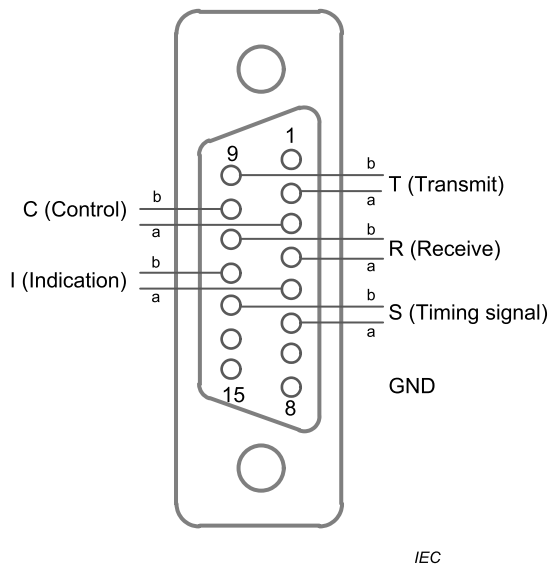


Figure 7 – Commonly used V.11 connector

For the physical characteristics (voltage, connector form factor, insulation, and pin-out) of these interfaces the respective reference standards shall be applied.

4.4.2.3 Data link layer

Integrated FSK modems typically operate with a symbol rate between 50 Baud and 1 200 Baud.

The corresponding reference standards are given in Table 1.

Table 1 – FSK symbol rate and related narrowband standards

FSK symbol rate	Related narrowband standard
50 Baud	ITU-T R.35
100 Baud	ITU-T R.37
200 Baud	ITU-T R.38A
200 Baud	ITU-T R.38B
600 Baud	ITU-T V.23
1 200 Baud	ITU-T V.23

Integrated voice frequency modems according to ITU-T use a frequency band between 300 Hz and 3 400 Hz and achieve data rates of 14,4 kbps or higher. However, these standards have not been designed for use over PLC. At start-up, after noise bursts or short channel interruptions, and for compatibility to earlier ITU-T standards, the modems go through a training sequence with a duration of several seconds before data transmission can start or continue.

4.4.2.4 Ethernet IEEE 802.3 interface

4.4.2.4.1 General

In some cases it is requested to interface APLC links with a local area network LAN in order to implement local switching or bridging functionalities.

This integrated interface often implements both the Ethernet 10/100BaseT (copper) as well as 100FX (optical) standard interfaces.

Ethernet frames are first serialized, elaborated and then transmitted through an internal integrated high speed data serial modem of the APLC terminal.

4.4.2.4.2 Physical layer

For Ethernet interfaces twisted copper pair or optical connectors are commonly used. Possible examples for these connectors are given in Figures 8 and 9.

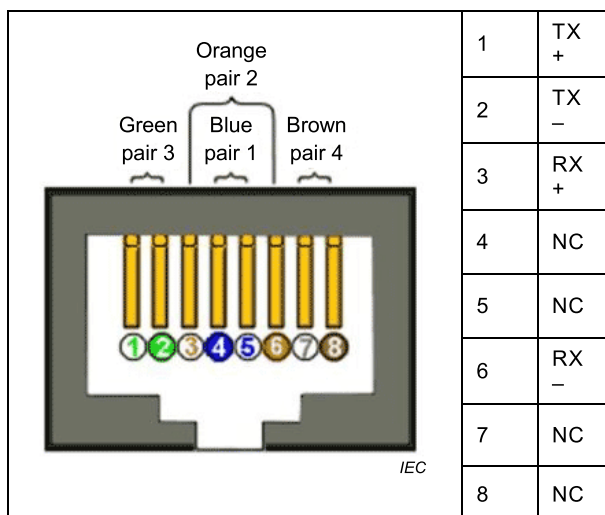


Figure 8 – ETH IEEE 802.3 RJ45 type connector

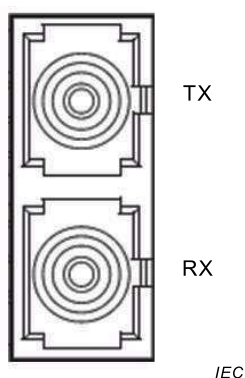


Figure 9 – ETH IEEE 802.3 SC type connector

For the physical characteristics (voltage, connector form factor, insulation, and pin-out) of these interfaces the respective reference standard shall be applied.

4.4.2.4.3 Data link layer

For this application the link layer shall be compliant with the Ethernet standard IEEE 802.3.

For practical reasons more than one Ethernet port shall be made available implementing switching functionalities for LAN.

Remote Ethernet connectivity requests, at least, the implementation of the basic transparent WAN bridging functionalities according to IEEE 802.1D.

5 Transmission line side high frequency interface

5.1 APLC high frequency band and channelling

The typical high frequency range used in Europe spans from 40 kHz to 500 kHz (up to 1 000 kHz in some countries and for special applications (e.g. cables) down to 24 kHz). Parts of this range may be barred by national regulations.

The nominal high frequency band is divided into basic high frequency bands of typically 4 kHz or 2,5 kHz according to the used LF channels.

Generally an APLC terminal operates in a nominal high frequency band consisting of one or more adjacent basic high frequency bands for each direction of the transmission.

The nominal high frequency band of the terminal has a bandwidth equal to $n \times B$ (n = number of high frequency channels; B = bandwidth of basic high frequency band).

The nominal transmit and receive high frequency bands of an APLC terminal may be non-adjacent or adjacent.

For each APLC terminal, the configurable set of high frequency nominal channel bandwidths and the set of nominal carrier frequencies for which the recommended values will be maintained shall be stated.

5.2 Frequency accuracy

The high frequency, corresponding to the carrier possibly suppressed during the modulation process, for any high frequency channel shall not differ from its nominal value by more than ± 10 Hz.

5.3 Signal levels

The configurable range of PEP and RMS power of the high frequency signal at the transmission line side high frequency interface when loaded with nominal impedance shall be stated in the terminal user manual.

For the power distribution over the nominal HF band, Annex C of this document and Annex C of IEC 62488-1:2012 may serve as guidelines.

5.4 Nominal impedance

The nominal impedance within the nominal high frequency transmit band at the transmission line side high frequency interface shall be 75 Ω unbalanced or 150 Ω balanced.

Other values may be considered if needed. For each value, the HF interface of the terminal shall comply with the requirements defined in Clause 5.

5.5 Return loss

The test procedure for measuring the return loss is described in 7.3.

The return loss within the nominal high frequency transmit band shall be not less than 10 dB.

5.6 Degree of unbalance to earth

The test procedure for measuring the degree of unbalance to earth is described in 7.4.

For the transmission line side high frequency interface of the balanced type the longitudinal conversion loss shall be not less than 40 dB at power frequency 50 Hz or 60 Hz.

5.7 Tapping loss

The test procedure for measuring the tapping loss is described in 7.5.

The impedance outside the band B_0 depicted in Figure 10 shall be such that any other APLC or DPLC terminal, connected to the same coupling matching unit with the same impedance, shall not suffer from a tapping loss A_t higher than indicated in Figure 10.

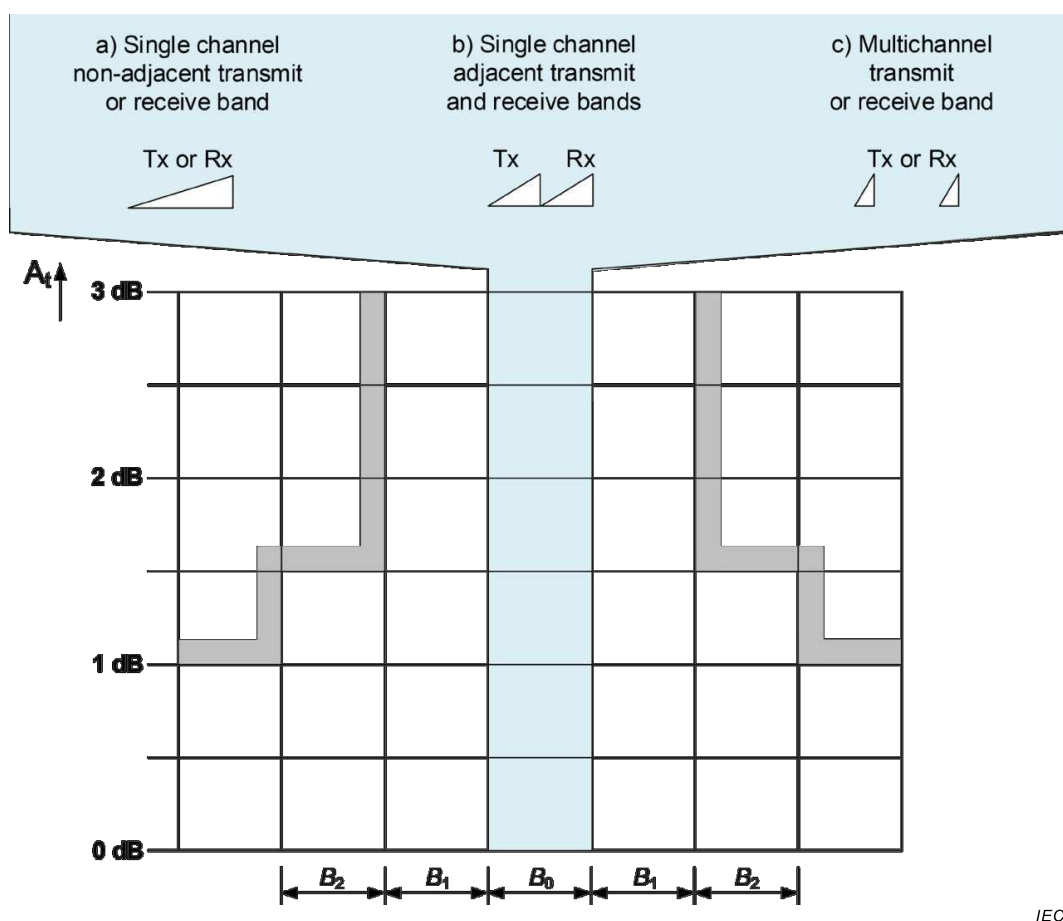


Figure 10 – Tapping loss limits for APLC terminals

The values of B_0 , B_1 , and B_2 are given in the table below where B is the basic high frequency band (typically 4 kHz or 2,5 kHz) and n is the number of channels.

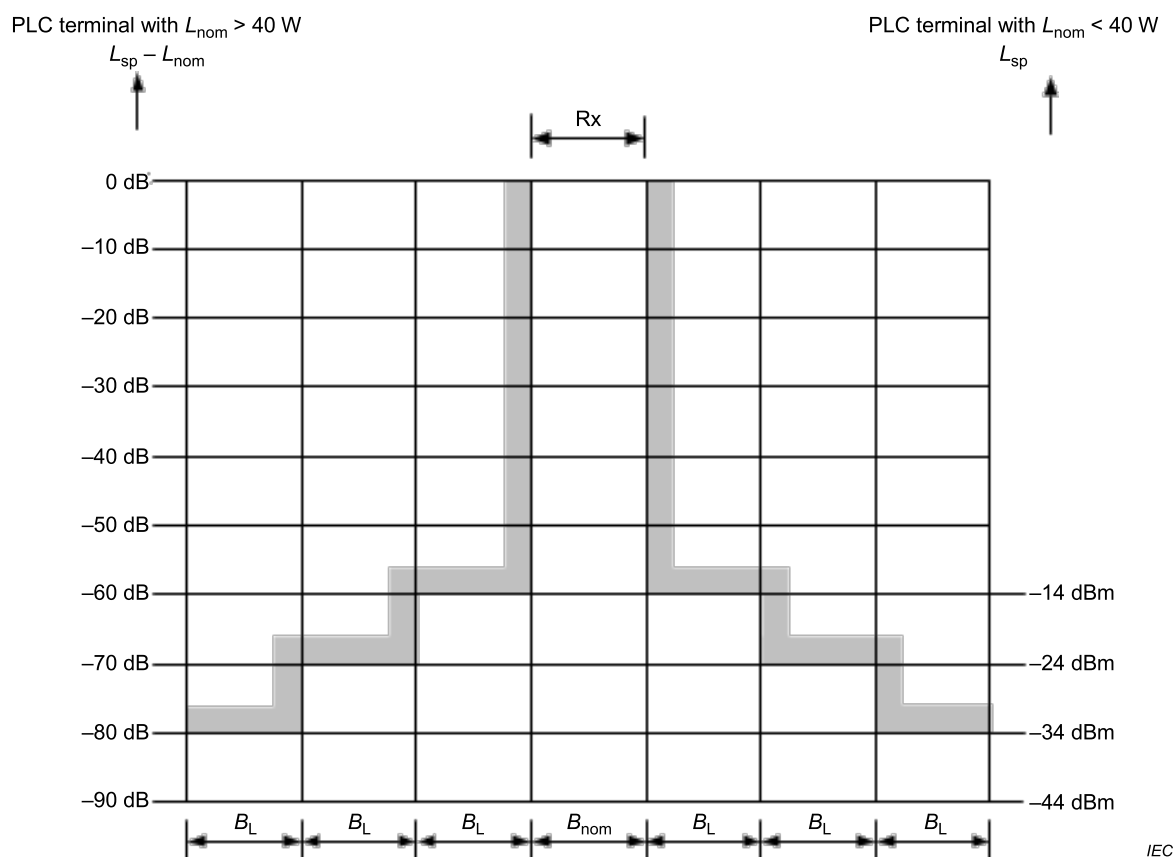
	B_0	B_1	B_2
a)	B	$2 \times B$	B
b)	$2 \times B$	$2 \times B$	B
c)	$n \times B$	$n \times B$	$n \times B$

5.8 Spurious emissions

Spurious emissions are emissions, at one or more frequencies, located outside the nominal high frequency transmit band.

The test procedure for measuring the spurious emissions is described in 7.6.

The maximum permitted level of spurious emissions is indicated in Figure 11.



B_{nom} is the bandwidth of the nominal high frequency band used for transmission.

$B_{nom} = n \times B$ where B is the bandwidth of the basic high frequency band and n the number of channels.

L_{sp} is the level of spurious emissions

L_{nom} is the nominal high frequency band output power

$B_L = B_{nom}$

Figure 11 – Maximum level of spurious emissions outside the high frequency band

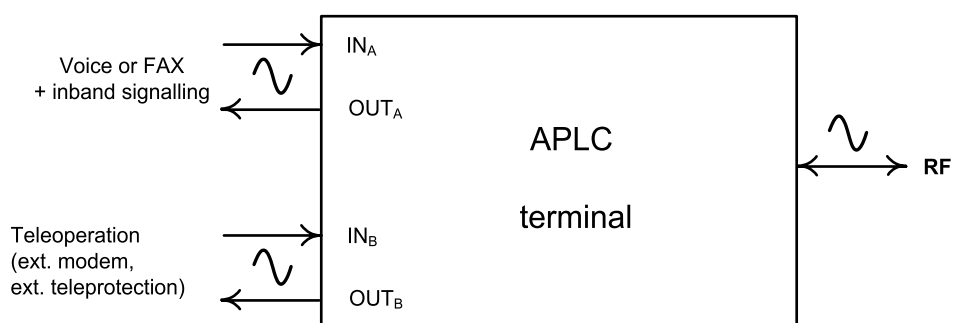
6 Quality and Performance

6.1 General

In Clause 6, several requirements concerning the quality and performance of a single or a couple of interconnected APLC terminals will be given.

The requirements particularly focus on APLC requirements referring to both the voice frequency side interfaces as well as the high frequency side interface for a single or a pair of terminals simulating an APLC link.

Figure 12 shows the reference points for measuring APLC parameters. These are the two analogue low frequency interfaces appearing in Figure 2.



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Figure 12 – Reference points for measuring APLC parameters

6.2 APLC internally generated noise

The none-weighted noise level, measured at the voice-frequency output of a pair of PLC terminals shall not exceed $-52,4 \text{ dBm0}$. This value can be considered equivalent to a weighted noise level of -55 dBm0p .

The measurement shall be made in the absence of any signal transmission (except for pilot signal or reduced carrier).

6.3 Automatic gain control

In the case of a 30 dB change in high frequency received signal level within the regulation range, the change in voice-frequency receive levels of both speech and signals shall be less than 1 dB.

The automatic gain control characteristic shall be coordinated with the teleprotection requirements.

6.4 Limiter action

The limiter associated with the voice channel shall produce a limiting effect defined as follows.

The action of the limiter shall start between -3 dBm0 and 0 dBm0 for any sinusoidal signal of a frequency between 300 Hz and the upper frequency of the voice channel.

The level at the high frequency side interface of the output signal measured by means of a true RMS non-selective level meter shall not exceed $+3 \text{ dBm0}$ for an increase in the voice-frequency input signal to a level of $+15 \text{ dBm0}$.

When the APLC operation is not degraded at the maximum output level as a result of a link budget design, a limiter is not necessary.

6.5 Transmit/Receive frequency difference

In a pair of APLC terminals, the frequency difference between a voice-frequency signal applied to the transmit end and that received at the receive end shall not exceed 2 Hz.

For data transmission automatic frequency control of the APLC terminals is recommended.

6.6 Attenuation distortion

The attenuation distortion of a pair of APLC terminals measured between the voice input and output (points IN_A and OUT_A in Figure 12) shall meet the requirements in Figures 13, 14 and 15.

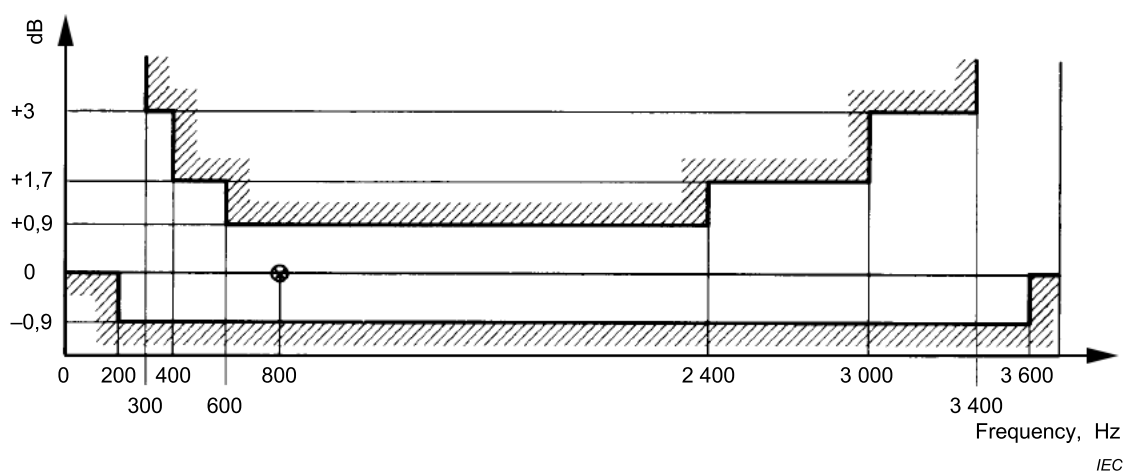


Figure 13 – Attenuation distortion limits for the voice frequency band of 300 Hz to 3400 Hz (ITU-T G.232)

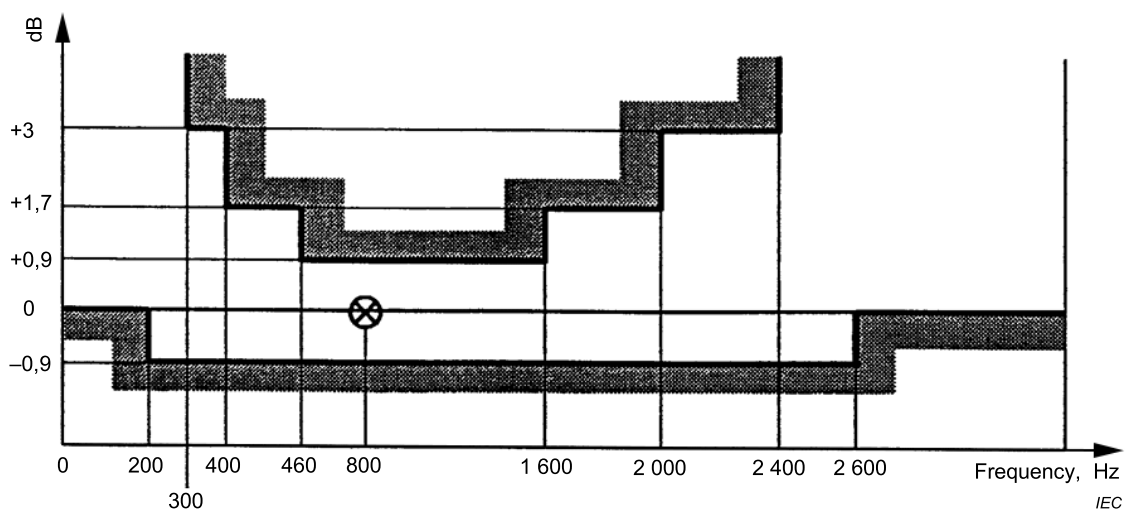


Figure 14 – Attenuation distortion limits for the voice frequency band of 300 Hz to 2400 Hz

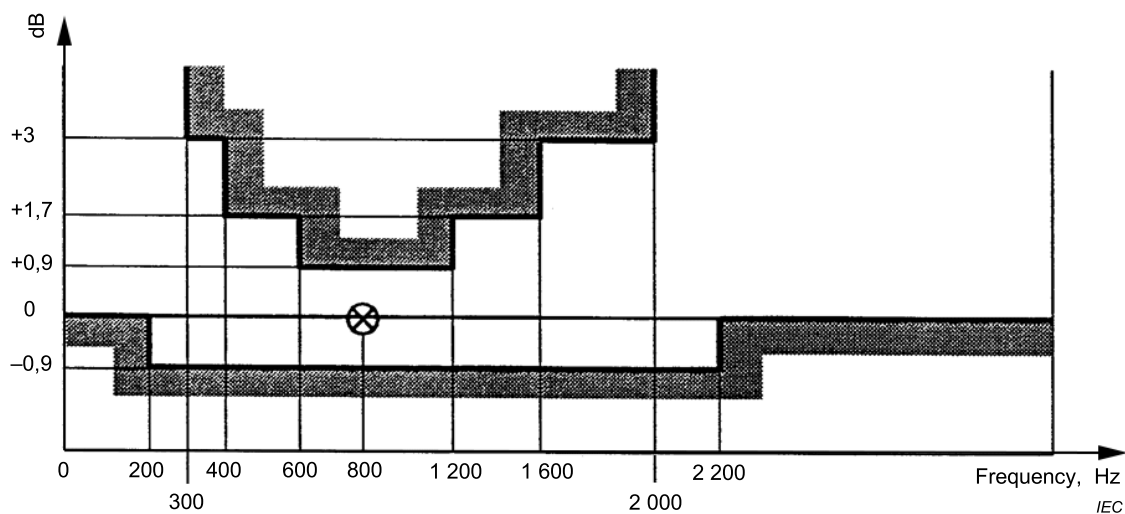


Figure 15 – Attenuation distortion limits for the voice frequency band of 300 Hz to 2000 Hz

6.7 Group-delay distortion

The group-delay distortion of a pair of APLC terminals measured between the voice-frequency input and output (points IN_A and OUT_A in Figure 12) shall meet the requirements in Figures 16, 17 and 18.

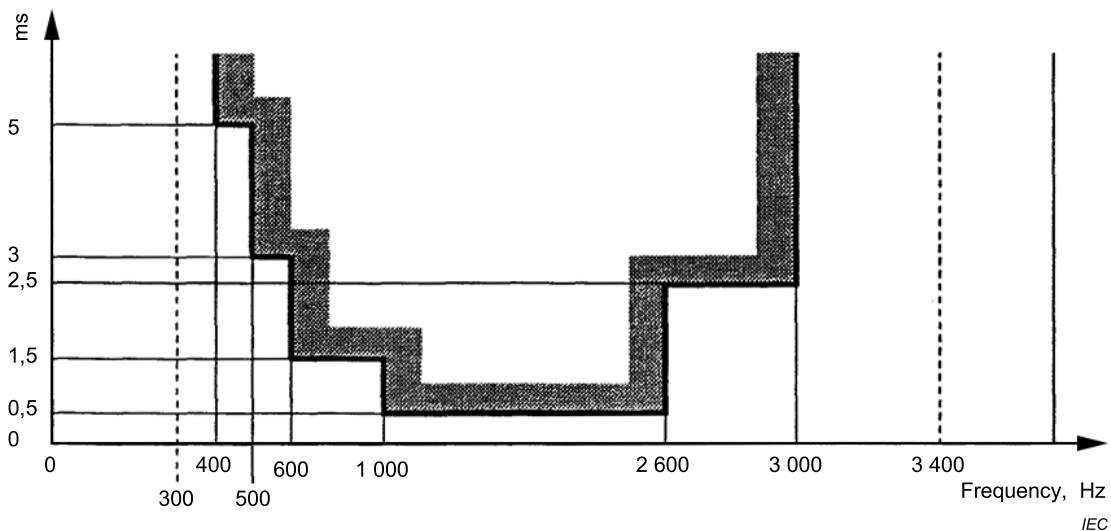


Figure 16 – Group delay distortion limits for the voice frequency band of 300 Hz to 3400 Hz

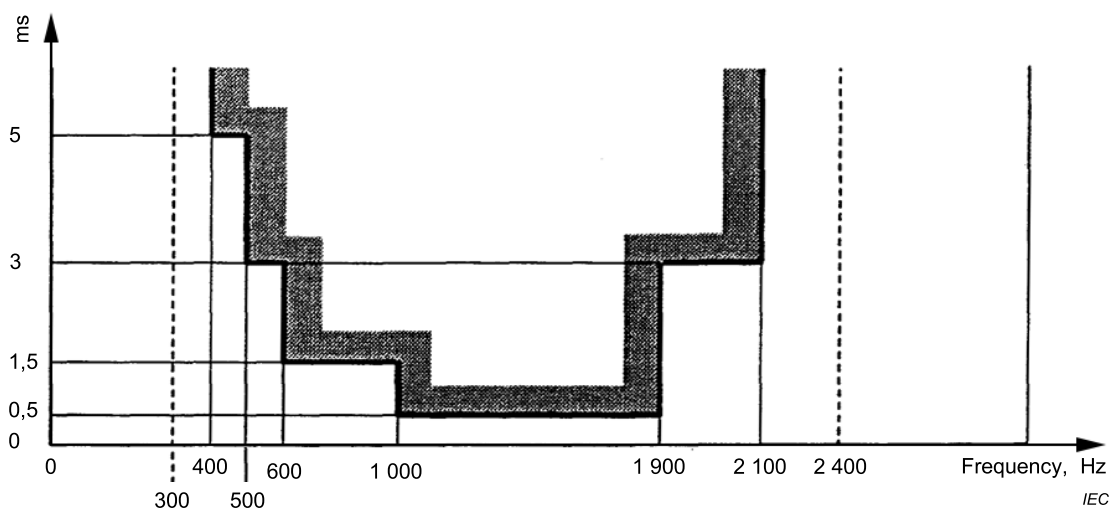


Figure 17 – Group delay distortion limits for the voice frequency band of 300 Hz to 2400 Hz

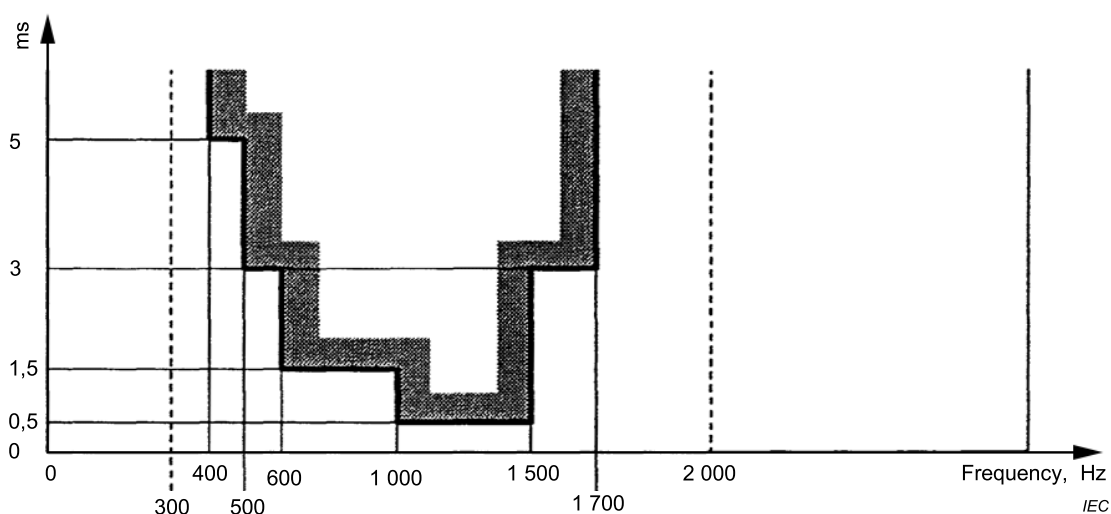


Figure 18 – Group delay distortion limits for the voice frequency band of 300 Hz to 2000 Hz

6.8 Harmonic distortion

The harmonic distortion of a pair of APLC terminals shall be measured at the output (point OUT_A in Figure 12).

A sinusoidal test signal with a frequency of 350 Hz and a level of -3 dBm0 shall be injected at the input (point IN_A in Figure 12).

The level of each component present at the voice frequency output (point OUT_A in Figure 12) shall be lower than -40 dBm0.

6.9 Selectivity

The test arrangement is described in 7.7.

When a sinusoidal signal with a frequency of 300 Hz outside the nominal carrier-frequency receive band is injected at the carrier-frequency side of a PLC terminal with a level of $+10$ dBm0, the level of the received signal at the voice-frequency side (points OUT_A or OUT_B in Figure 12) shall be less than -55 dBm0.

For a basic high frequency band of B, when a sinusoidal signal with a frequency of B or more outside the nominal high frequency receive band is injected with a level of $+20$ dBm0, the level of the received signal at the voice-frequency side (points OUT_A or OUT_B in Figure 12) shall be less than -55 dBm0.

6.10 Crosstalk attenuation

6.10.1 Co-channel crosstalk attenuation

The test arrangement is described in 7.8.

The near-end crosstalk attenuation measured between transmitting points IN_A or IN_B of a channel and receiving points OUT_A or OUT_B of the same channel for the near-end terminal shall be not less than 50 dB.

6.10.2 Inter-channel crosstalk attenuation

The test arrangement is described in 7.8.

The near-end crosstalk attenuation measured between transmitting points IN_A or IN_B in Figure 12 of a channel and receiving points OUT_A or OUT_B of the other channels for the near-end terminal shall be not less than 50 dB.

The far-end crosstalk attenuation measured between receiving points OUT_A or OUT_B in Figure 12 of the considered channel and receiving points OUT_A or OUT_B of the other channels for the far-end terminal shall be not less than 50 dB.

7 Testing

7.1 General

In Clause 7 a description of the test setup and methodology to be applied for assessing the main features of an APLC terminal will be given. All tests shall be performed in order to measure and verify that meet the requirements expressed in Clauses 4, 5 and 6.

All the tests verifying the requirements shall be considered as type tests as defined in the International Electrotechnical Vocabulary (IEV 151-04-15, IEC 60050).

7.2 Test setup for APLC link tests

In all test procedures which require the use of a pair of APLC terminals, the high frequency side interfaces shall be connected by an artificial line (resistive attenuator with attenuation A), matched to the nominal impedance of the terminals according to Equation (4).

$$A = 20\text{dB} \quad (4)$$

As modern electronic components allow the conception of automatic gain control circuits with a wide regulation range, it is assumed that a 20 dB attenuation is suitable for testing any PLC in a range of power from 1 W to 50 W. However, the attenuation value can be different, if required.

7.3 Return loss

This paragraph describes the test procedure for measuring the return loss. The return loss of the high frequency interface of an APLC terminal shall be measured to verify that it meets the requirement defined in 5.5.

The return loss measurement shall also be performed on all the voice frequency interfaces of the APLC terminal analogue low frequency side interface.

The test circuit shown in Figure 19 can be used.

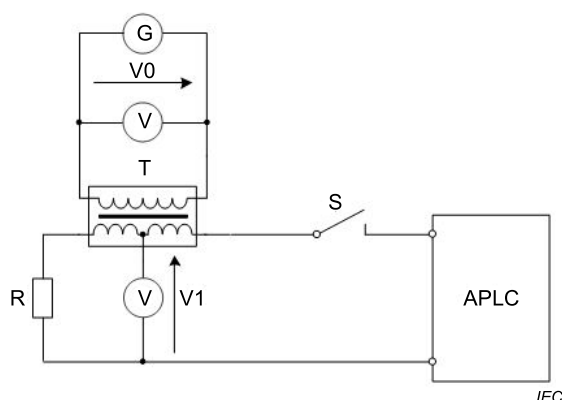


Figure 19 – Test circuit for return loss measurement

The resistor R shall be equal to the nominal impedance Z_{nom} of the APLC interface under test. That is generally 600 Ω for the audio frequency interfaces, and 75 Ω or 150 Ω for the high frequency interface.

G is a sine wave generator, and V0 shall be a sinusoidal signal with a constant voltage throughout the specified frequency range. The voltage level shall be the same in both positions of the switch S. A voltage of 1 V_{rms} is recommended.

T is a symmetric differential transformer wound around a magnetic material suitable for the given frequency band. It is composed of two windings, the primary on the generator side, the secondary, center tapped, on the APLC side. In principle, the transformer ratio is 1:1 primary to secondary (center tapped).

The measurement of voltages V0 and V1 shall be carried out with a suitable frequency selective voltmeter with high impedance across the specified frequency range. The APLC is in idle state with no signal transmitted.

For each frequency, the measurement is carried out with switch S in open position giving the voltage $V1_{open}$ and with S in closed position giving the voltage $V1_{closed}$.

The return loss A_{RL} is then given by Equation (5):

$$A_{RL} = 20 \times \log_{10} \left| \frac{V1_{open}}{V1_{closed}} \right| \text{ dB} . \quad (5)$$

Note on the measurement with the differential transformer:

As the elements of the measurement circuit are not perfect, the value of V0 may change slightly when closing/opening the switch S, and may also change from one frequency to another. It is then mandatory to adjust and verify V0 at each point. This can be avoided using a transformer with a turn ratio of 10 for example. This shall mask the variation of the impedance of the secondary side seen from the generator side.

7.4 Degree of unbalance to earth

7.4.1 General

This subclause describes the test procedure for measuring the Longitudinal Conversion Loss (LCL) and the Output Signal Balance (OSB) which are related to the degree of unbalance to earth.

The LCL shall be measured on all the voice frequency channels inputs (including telegraph and teleprotection transmission ports) within the voice frequency transmitted band.

It shall also be measured on the balanced high frequency interface at mains power frequency 50 Hz or 60 Hz.

The OSB shall be measured on all the voice frequency channels outputs (including telegraph and teleprotection receiving ports) within the voice frequency transmitted band.

7.4.2 LCL

The test circuit is given in Figure 20 (other arrangements may be considered as far as they follow the ITU-T Recommendation O.9).

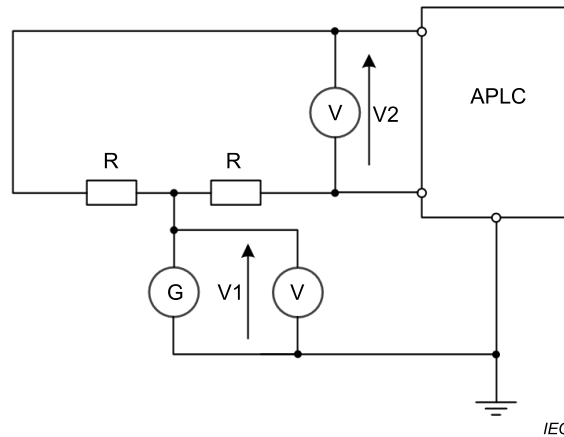


Figure 20 – Test circuit for LCL measurement (Tx port)

The resistors R shall be $Z_{nom}/2$ (that is generally $R = 300 \Omega$ for the voice frequency interface and $R = 75 \Omega$ for the balance high frequency interface).

G is a sine wave generator, and V1 shall be a sinusoidal signal with a constant voltage throughout the specified frequency range which is:

- 50 Hz to 3 400 Hz for the voice frequency channel inputs
- 50 Hz to 60 Hz for the high frequency interface

A voltage of 1 V_{rms} is recommended. the voltages V1 and V2 measurement shall be carried out with a suitable frequency selective voltmeter with impedance $\gg R$ across the specified high frequency range.

The APLC terminal is in idle state with no signal transmitted.

The measured values of V1 and V2 are used to calculate the LCL by using Equation (6) at each measurement point:

$$LCL = 20 \times \log_{10} \left| \frac{V1}{V2} \right| \text{ dB} \quad (6)$$

7.4.3 OSB

The test circuit is given in the Figure 21 (other arrangement may be considered as far as they follow the ITU-T Recommendation O.9).

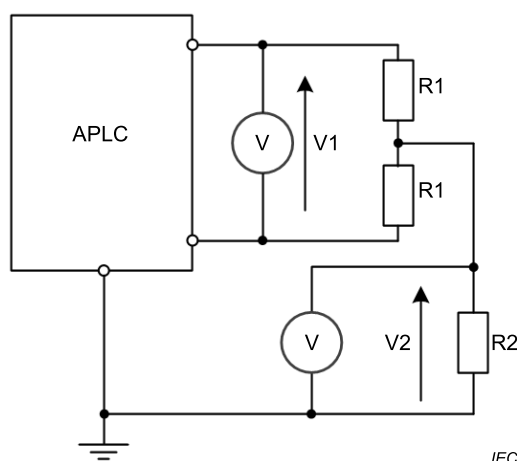


Figure 21 – Test circuit for OSB measurement (Rx port)

The resistors R1 shall be $Z_{\text{nom}}/2$ and the resistor R2 shall be $Z_{\text{nom}}/4$ (that is generally $R1 = 300 \, \Omega$ and $R2 = 150 \, \Omega$).

The voltages V1 and V2 measurement shall be carried out with a suitable frequency selective voltmeter with impedance $\gg R$ across the specified frequency range (50 Hz to 3 400 Hz). The APLC is in idle state with no signal transmitted.

The measured values of V1 and V2 are used to calculate the LCL by using Equation (7) at each measurement point:

$$OSB = 20 \times \log_{10} \left| \frac{V1}{V2} \right| \text{ dB} \quad (7)$$

7.5 Tapping loss

Subclause 7.5 describes the test procedure for measuring the Tapping loss. The test circuit is given in Figure 22. The closing of switch S simulates the insertion of the APLC into a matched circuit.

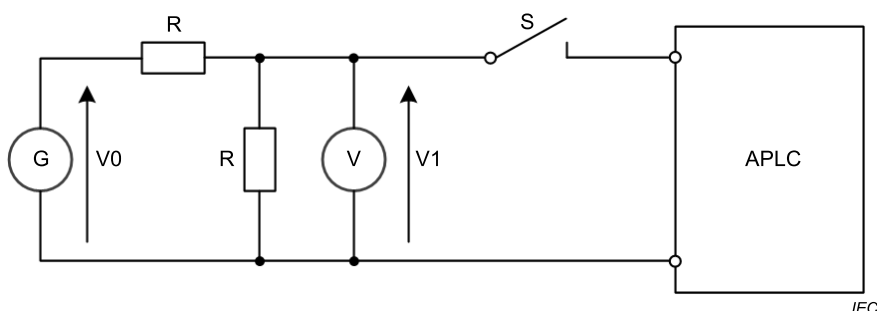


Figure 22 – Test circuit for Tapping Loss measurement

The resistors R shall be equal to the nominal impedance of the APLC high frequency interface Z_{nom} .

G is a sine wave generator, and V0 shall be a sinusoidal signal with a constant voltage of 775 mV RMS throughout the specified voltage range.

The measurement of voltage V_1 shall be carried out with a suitable frequency selective voltmeter with impedance $\gg R$ across the specified frequency range. The APLC is in idle state with no signal transmitted.

For each frequency, the measurement is carried out with switch S1 in open position giving the voltage $V_{1\text{open}}$ and with S1 in closed position giving the voltage $V_{1\text{closed}}$. The Tapping Loss attenuation A_{TL} is then given by equation (8):

$$A_{\text{TL}} = 20 \times \log_{10} \left| \frac{V_{1\text{open}}}{V_{1\text{closed}}} \right| \text{ dB} \quad (8)$$

7.6 Spurious emissions

7.6.1 Single channel terminals

For single channel terminals, in order to measure spurious emissions, the transmitter shall be modulated by two sinusoidal test signals of equal amplitude, the transmitter being terminated with a resistive load equal to the nominal impedance.

For APLC with a channel bandwidth B of 4 kHz or more the test signal frequencies shall be 350 Hz and $B - 700$ Hz.

For APLC with a channel bandwidth B of 2,5 kHz or less the test signal frequencies shall be 350 Hz and $B - 200$ Hz.

For APLC with a channel bandwidth between 2,5 kHz and 4 kHz the test signal frequencies shall be 350 Hz and $B - 400$ Hz.

The amplitude of the test signals shall be chosen so that each produces a quarter of the stated nominal high frequency output power.

The measurement shall be made with the aid of a selective level measuring set with an adequate bandwidth. It is necessary to ensure that the limiting action of the limiter, if any, does not occur when making this measurement.

7.6.2 Multi-channel terminals

For multi-channel terminals, in order to measure spurious emissions, the transmitter shall be modulated by a sinusoidal voice-frequency signal applied simultaneously with the same amplitude to each voice-frequency channel interface, the transmitter being terminated with a resistive load equal to the nominal impedance.

The frequency of the test signal shall be $B/2$, where B is the lowest channel bandwidth.

The amplitude of the test signal shall be chosen so as to produce a $1/n^2$ part of the stated nominal carrier-frequency output power of the terminal, where n is the number of channels.

The measurement shall be made with the aid of a selective level measuring set with an adequate bandwidth. It is necessary to ensure that the limiting action of the limiter, if any, does not occur when making this measurement.

7.7 Selectivity

The test is carried out on a link by injecting the signal at the high frequency side interface of one terminal by means of an external signal source and a hybrid using a test circuit as shown in Figure 23.

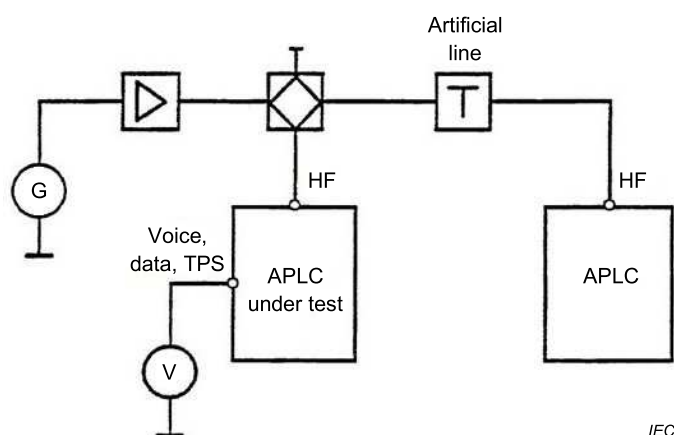


Figure 23 – Test circuit for selectivity measurement

7.8 Co-channel and inter-channel crosstalk attenuation

The requirements apply to a pair of APLC terminals forming an APLC link.

The test shall be carried out by injecting two sinusoidal voice frequency signals in each channel in the absence of any other signal transmission.

The amplitude of these test signals shall be chosen so that each one produces a $1/(4n^2)$ part of the stated nominal high frequency output power of the APLC terminal, where n is the number of channels.

For systems with a channel bandwidth B of 4 kHz or more the signal frequencies shall be 350 Hz and $B - 700$ Hz.

For systems with a channel bandwidth B of 2,5 kHz or less the signal frequencies shall be 350 Hz and $B - 200$ Hz.

For systems with a channel bandwidth between 2,5 kHz and 4 kHz the signal frequencies shall be 350 Hz and $B - 400$ Hz.

The measurement shall be made with a selective level measuring set with an adequate bandwidth and shall include harmonics and inter-modulation products. Limiting shall not take place when making this measurement.

8 Configuration and management

8.1 General

Basically every APLC terminal using the DSP technology has its own supervisory and programming interface. It allows the configuration and management of all main parts composing the terminal.

8.2 Configuration

A web based interface or software application shall be made available for the local configuration and the execution of the key maintenance operations.

It can be used to configure the main APLC terminal parameters in software mode as well as bandwidth, transmission and reception frequency, transmission power, user services, teleprotection services.

However, in order to complete the terminal configuration, certain hardware pre-settings have to be adjusted.

In addition to the above features in order to simplify the APLC link setup it shall be useful to have a voice channel to be used during configuration steps. This enables operators at the two ends of the link to talk during the configuration phase.

8.3 Network management system

The network management system allows diagnostics and maintenance of the APLC terminals from a remote control center.

A dedicated software, which is out of the scope of this standard, allows the user through configuration and maintenance remote operations to manage APLC terminals and links by an easy-to-understand interface display.

It is therefore often requested to integrate APLC management function into a generic SNMP (Simple Network Management Protocol) manager. In this case the APLC terminal MIB (Management Information Base) shall be made available.

In particular to make available access to the MIB information or execute a command, sending alarm traps to SNMP manager, APLC terminals shall use a network management system interface.

8.4 Local terminal alarms

DSP ensure a very high level of reliability and in case of any APLC terminal cards or line failure the internal diagnostic routine shall return to the operator an alarm, displayed on the front of the APLC terminal or signaled by dry relay contacts.

APLC terminal shall have on board a number of LED indicators that will summarize the status of operation of the APLC terminal and/or the link highlighting the proper link operation or the presence of anomalies/failures.

Whenever APLC enters a wrong operation an alarm occurs. According to the alarm returned at least two different type of conditions shall be shown. Commonly used are:

	Alarm type	Meaning
1	Transmitter alarm	A hardware error has been detected in the transmitter and/or the transmitted signal is missing or low
2	Receiver alarm	A hardware error has been detected in the receiver and/or the received signal is low or distorted
3	Hardware alarm	A hardware error has been detected in the local terminal
4	Link alarm	The received signal is low or distorted

9 Cyber security

9.1 General

The configuration and monitoring through the network software application raises problems relating security and confidentiality of the information exchanged between each APLC terminal and the remote control center. Several aspects concerning cyber security are well addressed in the IEC 62351 series.

For APLC at least the following technology must be considered:

- User activity logging. This is a way to trace any activity performed on the terminal through its configuration software application;
- Traffic encryption, so it cannot be intercepted, freely analyzed and eventually modified by a third party;
- Implementing authentication mechanism that prevents a third party generating false messages without them being recognized as not valid.

9.2 Authentication

Authentication is the mechanism allowing to limit the terminal access to users having the necessary privileges only; basically there are two types of authentication mechanisms – local and remote – as in the following:

- The local authentication mechanism evaluates the details supplied by the user, comparing them with those recorded on the APLC terminal; there are two different user profiles (i.e. root and guest, respectively) that defines the different levels of privilege.
- The local access password can be modified via the local APLC terminal configuration software.
- The remote authentication mechanism sends authentication requests received at the APLC terminal input to a remote authentication server which use a secret code shared between the terminal and the server for authenticating the access requests. The server authenticates the user and provides a positive or negative feedback on whether the details provided by the user are correct or not.

All necessary parameters for these authentication modes can be usually defined through the APLC terminal configuration software.

10 APLC safety

10.1 General

Clause 10 specifies the requirements for the terminal in order to reduce the risk of fire, electric shock or injury to the user. It is assumed that access to the terminal in normal service is restricted to personnel aware of working procedures to ensure safety.

Clause 10 does not cover the functional safety requirements and it does not cover the safety requirements of installations.

10.2 Safety reference standard

The safety reference standard for the APLC terminal is IEC 60255-27. As the scope of this standard is the safety requirements for protection devices, it applies to terminals which will normally be installed in a restricted access area within a power station, substation or industrial/retail environment. It is therefore an obvious choice to extend its application to the APLC terminal safety requirements.

Alternatively IEC 60950-1 may be used to prove the conformity of APLC regarding safety.

10.3 Classification of APLC Terminals

The APLC terminal shall meet the requirements of Table C.6 of IEC 60255-27:2013 and Table C.10 of IEC 60255-27:2013, as shown in Tables 2 and 3.

Table 2 – Basic insulation [Table C.6 of IEC 60255-27:2013]

Nominal rated insulation voltage or working voltage (RMS or DC) up to									
	Clearance	Creepage distance					Withstand voltage ^c		
		mm					V		
		In equipment			On printed wiring board		Peak impulse	RMS. 50/60 Hz	DC
		Material group			Not coated	Coated ^{a b}			
V	mm	I CTI ≥ 600	II CTI ≥ 400	III CTI ≥ 100	CTI ≥ 175	CTI ≥ 100	1,2/50 µs	1 min	1 min
50	0,15	0,60	0,85	1,20	0,15	0,10	800	490	700
100	0,50	0,70	1,00	1,40	0,50	0,50	1 500	820	1 150
150	1,50	1,50	1,50	1,50	1,50	1,50	2 500	1 350	1 900
300	3,00	3,00	3,00	3,00	3,00	3,00	4 000	2 200	3 100
600	5,50	5,50	5,50	6,00	5,50	5,50	6 000	3 250	4 600
1 000	8,00	8,00	8,00	10,00	8,00	8,00	8 000	4 350	6 150

^a Also applies to PCB-mounted components with mechanically stable distances between leads.

^b See C.2.2 of IEC 60255-27:2013 for minimum coating requirements.

^c For proving the clearance in air, see 10.6.3 and Table C.11 of IEC 60255-27:2013.

Linear interpolation may be used between the nearest two points, the calculated minimum clearance and creepage distance being rounded up to the next higher specified increment. The specified increment is 0,1 mm.

Table 3 – Double or reinforced insulation [Table C.10 of IEC 60255-27:2013]

Nominal rated insulation voltage or working voltage (RMS or DC) up to	Pollution degree 2 – Overvoltage category III								
	Clearance	Creepage distance					Withstand voltage ^c		
		mm					V		
		In equipment			On printed wiring board		Peak impulse	RMS. 50/60 Hz	DC
		Material group			Not coated	Coated ^{a b}			
V	mm	I CTI ≥ 600	II CTI ≥ 400	III CTI ≥ 100	CTI ≥ 175	CTI ≥ 100	1,2/50 µs	1 min	1 min
50	0,50	1,20	1,70	2,40	0,50	0,50	1 500	820	1 150
100	1,50	1,50	2,00	2,80	1,50	1,50	2 500	1 350	1 900
150	3,00	3,00	3,00	3,00	3,00	3,00	4 000	2 200	3 100
300	5,50	5,50	5,50	6,00	5,50	5,50	6 000	3 250	4 600
600	8,00	8,00	11,00	12,00	8,00	8,00	8 000	4 350	6 150
1 000	14,00	14,00	14,00	20,00	14,00	14,00	12 000	6 500	9 200

^a Also applies to PCB-mounted components with mechanically stable distances between leads.

^b See C.2.2 of IEC 60255-27:2013 for minimum coating requirements.

^c For proving the clearance in air, see 10.6.3 and Table C.11. of IEC 60255-27:2013

Linear interpolation may be used between the nearest two points, the calculated minimum clearance and creepage distance being rounded up to the next higher specified increment. The specified increment is 0,1 mm.

The following conditions shall be applied:

- pollution degree 2;
- overvoltage category III

as applicable for equipment operating in harsh environments of substations according to IEC 60255-27.

These conditions can be reduced if special measures are taken to limit transient voltages to appropriate values and/or additional protections are used to reduce the pollution degree.

In this case, these measures and protections shall be clearly described in the safety chapter of the terminal user manual.

10.4 Ingress protection

The terminal shall have an Ingress Protection equal or better than IP2x according to IEC 60529.

10.5 Type and routine tests

Table 4, which is taken from Table 12 of IEC 60255-27:2013, gives the list of the type and routine tests that are required to demonstrate the APLC terminal is compliant to the requirement.

The second column [Type Tests – Safety (Normative)] gives the minimum set of type tests to be carried out to demonstrate the compliance.

The third column gives the minimum set of tests that shall be carried out as routine test in factory on each single APLC terminal.

Table 4 – List of Type and Routine Tests [Table 12 of IEC 60255-27:2013]

Test	Reference	Type tests ^b		Routine test ^c
		Validation ^a (informative)	Safety (normative)	
Environmental tests				
Dry-heat test – operational	10.6.1.1	X	---	---
Cold test – operational	10.6.1.2	X	---	---
Dry-heat test at maximum storage temperature	10.6.1.3	X	---	---
Cold test at minimum storage temperature	10.6.1.4	X	---	---
Damp-heat test	10.6.1.5	X	---	---
Cyclic temperature with humidity	10.6.1.6	X	---	---
Vibration	10.6.2.1	X	---	---
Shock	10.6.2.2	X	---	---
Bump	10.6.2.3	X	---	---
Seismic	10.6.2.4	X	---	---
Reverse polarity and slow ramp	10.6.6	X	---	---
Safety-related tests				
Clearances and creepage distances	10.6.3	---	X	---
Accessible parts test	10.6.2.5	---	X	---
IP rating	10.6.2.6	---	X	---
Impulse voltage	10.6.4.2	---	X	---
AC or d.c. dielectric voltage	10.6.4.3	---	X	X
Insulation resistance	10.6.4.4	X	---	---
Protective bonding resistance	10.6.4.5	---	X	---
Protective bonding continuity	10.6.4.5.2	---	---	X
Flammability of insulating materials, components and fire enclosures ^d	10.6.5.2	---	X ^d	---
Single-fault condition	10.6.5.5	---	X	---
Electrical environment tests				
Maximum temperature of parts and materials	10.6.5.1	X	---	---
Thermal short time	10.6.5.3	X	---	---
Output relay, make and carry	10.6.5.4	X	---	---
^a Validation tests are normally carried out as type tests during product development but they may have an impact on product safety. After validation testing, the equipment under test should be checked for compliance with safety requirements, for example, due to cracking or distortion of parts providing insulation.				
^b Compliance with type-test requirements may be carried out by testing, measurement, visual inspection or assessment, as appropriate, for example, clearance and creepage distance measurement (or visual inspection where the distances are obviously large) or technical argument such as assessment of single-fault conditions where the result would be known. There shall be no electrical shock or fire hazard during or after conducting the normally applied type tests to demonstrate claimed compliance.				
^c For sample test guidance, refer to 10.6.4.3.2.3. of IEC 60255-27:2013				
^d Testing of plastic parts may be necessary where the material does not meet the minimum flammability specified in Clause 7 of IEC 60255-27:2013 or its thickness is below the minimum specified for that material to achieve the required minimum flammability.				
X applicable				
--- not applicable				

11 Storage and transportation, operating conditions, power supply

11.1 Storage and transportation

11.1.1 Climatic conditions

11.1.1.1 Storage

The terminal may be stored in partially open air locations and be partially exposed to limited wind-driven precipitation. Therefore the APLC terminal, together with its packaging and protection (not the terminal itself), shall meet the requirements of IEC 60721-3-1, class 1K5, as shown in Table 5.

Table 5 – Classification of climatic conditions [Table 1 of IEC 60721-3-1:1997]

[illegible]

Notes:

- 1) The low and high relative humidity's are limited by the low and high absolute humidity's, so that, for example, for environmental parameters a) and c), or b) and d), the severities given in Table 1 of IEC 60721-3-1:1997 do not occur simultaneously.
- 2) Averaged over a period of time of 5 min.
- 3) The value of 70 kPa represents a limit for open-air conditions, normally at an altitude of 3000 m. In some geographical areas, open-air conditions may occur at higher altitudes. Conditions in mines are not considered.
- 4) A cooling system based on non-assisted convection may be disturbed by adverse movement of surrounding air.
- 5) This rain temperature should be considered together with high air temperature b) and solar radiation j). The cooling effect of the rain has to be considered in connection with the surface temperature of the product.
- 6) These are air-conditioned locations with a tolerance of 2 °C on stated temperature value.
- 7) Conditions occurring at the location concerned to be selected from Table 2 of IEC 60721-3-1:1997.
- 8) If applicable, a special value may be selected from Table 2 of IEC 60721-3-1:1997.
- 9) Applies to wind-driven precipitation at partially weather protected locations.
- 10) The classes of climatic conditions of this document include the classes of IEC 60721-3-3 and IEC 60721-3-4 as follows:

1K1 covers 3K1	1K3 covers 3K5	1K5 covers 3K7	1K7 covers 4K1	1K9 covers 4K4	1K11 covers 4K6
1K2 covers 3K3	1K4 covers 3K6	1K6 covers 3K8	1K8 covers 4K2	1K10 covers 4K5	
- 11) Further information on classes 1K10 (tropical damp) and 1K11 (tropical dry) is given in Annex C of IEC 60721-3-1:1997.

For any worse conditions that may arise, for example in very cold or very wet countries, refer directly to IEC 60721-3-1:1997, A.2.1.

11.1.1.2 Transportation

The requirement is for the terminal, including protection (not the terminal itself) depends on the means of transportation and the transportation route.

IEC 60721-3-2 can serve as reference guide for transportation. Note that the conditions apply for the terminal together with its packaging, pallet and all the protections suitable for each particular means of transportation.

11.1.1.3 List of climatic tests for storage and transportation

Table 6 gives the minimum set of type tests to be carried out to demonstrate the conformity of the terminal.

Table 6 – Climatic tests for storage and transportation

Climatic Tests					
Test	IEC Standard		Temperature	Relative Humidity	Duration
Cold	IEC 60068-2-1	Test Ab	–40 °C	No	16 hours
Dry heat	IEC 60068-2-2	Test Bb	+70 °C	No	16 hours
Damp heat cyclic	IEC 60068-2-30	Test Db	+40 °C	95 %	(12 + 12 hours) x 2 cycles

11.1.2 Mechanical

The packing is of primary importance in the withstanding of the terminal to mechanical constraints during transportation and storage and shall be adapted to the different conditions that may be encountered.

The normal transportation conditions include all kinds of lorries and trailers in areas with a well-developed road system, transportation by planes, modern train system and ships. This corresponds to class 2M1 of the IEC 60721-3-2.

For any worse transportation conditions, for example transportation in areas without a well-developed road system, it is recommended to pack the terminal respecting class 2M2 of IEC 60721-3-2.

If testing is required, the following standards shall be applied:

- Sinusoidal vibrations (IEC 60068-2-6)
- Shocks (IEC 60068-2-27)
- Free fall and toppling (IEC 60068-2-31)

11.2 Operating conditions

11.2.1 Climatic conditions

Usually, APLC terminal equipment is installed in the telecom room building of the substation. This room may not always be air conditioned. In any case the room is equipped with a heating system. Consequently, the APLC terminal equipment shall meet the requirement of IEC 60721-3-3 class 3K5, amended as follows:

- **High air temperature:**
 - +45 °C
 - +55 °C for not more 24 hours per month without any degradation of performances.
- **Low air temperature:**
 - To prevent from a complete loss of APLC communication in case of failure of the heating system in very cold countries, the terminal shall operate without damage at temperatures down to –10 °C for a period of at least 16 hours per month without any degradation of performance and the lowest temperature for cold start shall be 0 °C.
- **Condensation, formation of ice:**
 - In normal operation the formation of ice shall not occur. The terminal equipment shall not be exposed to any type of water.
 - Temporary condensation may occur during maintenance when spare parts are introduced which have been stored at a lower temperature than that prevailing in the telecommunication terminal environment.

**Table 7 – Classification of climatic conditions from
Table 1 of IEC 60721-3-3:2002**

Environmental parameter	Unit	Class												
		3K1	3K2	3K3	3K4	3K5	3K6	3K7	3K7L	3K8	3K8H	3K8L	3K9 ⁸⁾	3K10 ⁸⁾
a) Low air temperature	°C	+20 ³⁾	+15	+5	+5	–5	–25	–40	–40	–55	–25	–55	+5	–20
b) High air temperature ⁵⁾	°C	+25 ³⁾	+30	+40	+40 ⁵⁾	+45 ⁵⁾	+55	+70	+40	+70	+70	+55	+40	+55
c) Low relative humidity	%	20	10	5	5	5	10	10	10	10	10	10	30	4
d) High relative humidity	%	75	75	85	95	95	100	100	100	100	100	100	100	100
e) Low absolute humidity	g/m ³	4	2	1	1	1	0,5	0,1	0,1	0,02	0,5	0,02	6	0,9
f) High absolute humidity	g/m ³	15	22	25	29	29	29	35	35	35	35	29	36	27
g) Rate of change of temperature ¹⁾	°C/min	0,1	0,5	0,5	0,5	0,5	0,5	1,0	1,0	1,0	1,0	1,0	1,0	1,0
h) Low air pressure ⁷⁾	kPa	70	70	70	70	70	70	70	70	70	70	70	70	70
i) High air pressure ²⁾	kPa	106	106	106	106	106	106	106	106	106	106	106	106	106
j) Solar radiation	W/m ²	500	700	700	700	700	1120	1120	None	1120	1120	1120	1120	1120
k) Heat radiation	None	No	6)	6)	6)	6)	6)	6)	6)	6)	6)	6)	6)	6)
l) Movement of surrounding air ⁴⁾	m/s	0,5	1,0 ⁵⁾	1,0 ⁵⁾	1,0 ⁵⁾	1,0 ⁵⁾	1,0 ⁵⁾	5,0 ⁵⁾	5,0 ⁵⁾	5,0 ⁵⁾	5,0 ⁵⁾	5,0 ⁵⁾	5,0 ⁵⁾	5,0 ⁵⁾
m) Condensation	None	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
n) Wind-driven precipitation (rain, snow, hail, etc.)	None	No	No	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
o) Water from sources other than rain	None	No	No	No	6)	6)	6)	6)	6)	6)	6)	6)	6)	6)
p) Formation of ice	None	No	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes

1) Averaged over a period of time of 5 min.

2) Conditions in mines are not considered.

3) These are air-conditioned locations with a tolerance of ± 2 °C on stated temperature values.

4) A cooling system based on non-assisted convection may be disturbed by adverse movement of surrounding air.

5) If applicable, a special value may be selected from table 2.

6) Conditions occurring at the locations concerned to be selected from table 2.

7) Severity value of 70 kPa covers worldwide application (altitudes up to 3 000 m). For some restricted applications, a value may be selected from table 2.

8) Further information on classes 3K9 (tropical damp) and 3K10 (tropical dry) is given in annex E.

11.2.2 Mechanical

The terminal is generally installed in the telecommunication room of electricity substations, where the vibrations are of low significance. Therefore the APLC terminal equipment shall meet the requirements of class 3M1, as specified in Table 6 of IEC 60721-3-3:2002 (see Table 8). Only when located very close to sources of high vibrations (e.g. generators within power stations) a higher level shall be applied.

**Table 8 – Classification of mechanical conditions from
Table 6 of IEC 60721-3-3:2002**

Environmental parameter	Unit	Class							
		3M1	3M2	3M3	3M4	3M5	3M6	3M7	3M8
a) <i>Stationary vibration, sinusoidal:</i>									
displacement amplitude	mm	0,3	1,5	1,5	3,0	3,0	7,0	10	15
acceleration amplitude	m/s ²	1	5	5	10	10	20	30	50
frequency range	Hz	2-9 9-200	2-9 9-200	2-9 9-200	2-9 9-200	2-9 9-200	2-9 9-200	2-9 9-200	2-9 9-200
b) <i>Non-stationary vibration including shock: (see figure 1)</i>									
shock response spectrum type L, peak acceleration $\hat{a}$	m/s ²	40	40	70	None	None	None	None	None
shock response spectrum type I, peak acceleration $\hat{a}$	m/s ²	None	None	None	100	None	None	None	None
shock response spectrum type II, peak acceleration $\hat{a}$	m/s ²	None	None	None	None	250	250	250	250

The following mechanical tests shall be carried out:

- Sinusoidal vibrations (IEC 60068-2-6)
- Shocks (IEC 60068-2-27)

Any other specific requirements for the mechanical environment, for example earthquake withstanding or installation in harsh industrial environment shall be taken into account at the level of the cabinet, not at the level of the APLC terminal itself.

11.2.3 Operating conditions set of tests

Tables 9, 10 and 11 give the minimum set of type test to be carried out to demonstrate the conformity of the terminal for climatic and mechanical conditions withstanding.

The way the tests have to be carried out, the performance criteria and the configuration of the terminal to be tested are described in the appropriate section of this document.

Table 9 – Climatic tests

Test	IEC Standard		Temperature	Relative Humidity	Duration
Cold (nominal) ¹⁾	IEC 60068-2-1	Test Ae	0 °C	No	24 hours
Cold (limit) ^{1), 2)}	IEC 60068-2-1	Test Ae	–10 °C	No	16 hours
Dry Heat	IEC 60068-2-2	Test Be	+55 °C	No	24 hours
¹⁾ If the test at the cold limit temperature is extended to duration of 24 hours, the test at the nominal temperature can be skipped. ²⁾ Cold start not required at –10°C					

Table 10 – Sinusoidal vibration test

Ref. IEC Standard	IEC 60068-2-6 – Test Fc. Both vibration response and vibration endurance ²⁾ tests shall be carried out		
Frequency range	Displacement	Acceleration	Axes
1 Hz – 10 Hz	0,3 mm	–	3 axes
10 Hz – 200 Hz	–	1 m/s ²⁾	3 axes
²⁾ Endurance by sweeping should be preferred, with 10 sweep cycles in each axis			

Table 11 – Non-repetitive shock test

Ref. IEC Standard	IEC 60068-2-27 – Test Ea		
Pulse Shape	Half-sine		
Peak Acceleration	40 m/s ²		
Duration	11 ms		
Number of shocks	18	3 successive shocks in each direction of 3 mutually perpendicular axes	

11.3 Power supply

11.3.1 AC supply

Nominal AC voltages shall be in accordance with the preferred values listed in IEC 60038.

The following values are the most common:

230 V ± 10 %	50 Hz or 60 Hz
120 V ± 10 %	60 Hz

The frequency tolerance shall be ±5 % and harmonic content <10 %.

11.3.2 DC supply

11.3.2.1 General

Nominal DC voltages shall be in accordance with the preferred values listed in standard IEC 60038. The following values are the most common.

24 V +20 %, –15 %
48 V +20 %, –15 %

The residual ripple peak-to-peak shall be ≤5 %.

The terminal shall be capable of operating in any of the following conditions:

- Positive pole earthed,
- Negative pole earthed,
- No reference to earth.

11.3.2.2 Reverse Polarity

Reverse polarity protection shall be provided in order to protect the terminal against the inadvertent inversion of the DC power supply voltage.

12 EMC

12.1 Emission and Immunity reference standards

The APLC terminal shall comply with the EMC requirements of IEC 61000-6-4 for emissions and the generic standard for equipment used in power station and substation environment IEC 61000-6-5 for immunity requirements.

It should be noted that IEC 61000-6-4 corresponds to the European standard EN 61000-6-4 which is a harmonized standard that can be used for conformity with the essential requirements of the EMC European Directive.

The requirements of the IEC 61000-6-2 standard series for industrial environments are covered by those of IEC 61000-6-5.

12.2 Emission

12.2.1 Radiated and conducted emission

Radiated and conducted emission limits are given in Tables 1, 2 and 3 of IEC 61000-6-4:2006 and IEC 61000-6-4:2006/AMD1:2010, as shown in Tables 12, 13 and 14.

Table 12 – Emission – Enclosure port [Table 1 of IEC 61000-6-4:2011 (ed.2.1)]

Table Clause	Port	Frequency range	Limits	Basic standard	Applicability note	Remarks
1.1	Enclosure Test facility: OATS or SAC	30 MHz to 230 MHz 230 MHz to 1 000 MHz	40 dB(μV/m) quasi-peak at 10 m 47 dB(μV/m) quasi-peak at 10 m	The measurement instrumentation shall be as defined in clause 4 of CISPR 16-1-1:2015. The measuring antennas shall be as defined in clause 4.4 of CISPR 16-1-4:2010. The measuring site shall be as described in clause 5 of CISPR 16-1-4:2010. The measurement method shall be as specified in clause 7.2 of CISPR 16-2-3:2016.	See a, b and e	May be measured at 30 m distance using the limits decreased by 10 dB. As stated in CISPR 16-2-3:2016 the antenna height shall be varied between 1 m to 4 m. Additional guidance on the test method can be found in CISPR 16-2-3:2016 clause 7.3 and clause 8.
1.2	Enclosure Test facility: FAR	30 MHz to 230 MHz 230 MHz to 1 000 MHz	52 dB(μV/m) to 45 dB(μV/m) quasi-peak at 3 m Limit reducing linearly with the logarithm of the frequency. 52 dB(μV/m) quasi-peak at 3 m	The measurement instrumentation shall be as defined in clause 4 of CISPR 16-1-1:2015. The measuring antennas shall be as defined in clause 4.4 of CISPR 16-1-4:2010. The measuring site shall be as described in clause 5.8 of CISPR 16-1-4:2010. The measurement method shall be as specified in clause 7.2.9.2 of CISPR 16-2-3:2016.	See a, b and e Only applicable to table top equipment	May be measured at greater distances with the limits decreased by 20 dB/decade (relative to distance) The limitations on EUT size in CISPR 16-1-4:2010 apply

Table clauses 1.2 and 1.3 are not relevant to APLC.

OATS Open area test site

SAC Semi-anechoic chamber

FAR Fully anechoic room

Table 13 – Emission – Low voltage AC mains port [Table 2 of IEC 61000-6-4:2011 (ed.2.1)]

Table clause	Port	Frequency range	Limits	Basic standard	Applicability note	Remarks
2.1	Low voltage AC mains	0,15 MHz to 0,5 MHz	79 dB(μV) quasi-peak 66 dB(μV) average	The measurement instrumentation shall be as defined in clauses 4 and 6 of CISPR 16-1-1: 2015. The measuring networks shall be as defined in clause 4 of CISPR 16-1-2: 2014. The measurement set up and method shall be as described in clause 7 of CISPR 16-2-1:2014.	See a and b	
		0,5 MHz to 30 MHz	73 dB(μV) quasi-peak 60 dB(μV) average			
^a Impulse noise (clicks) which occur less than five times per minute is not considered. For clicks appearing more often than 30 times per minute, the limits apply. For clicks appearing between 5 and 30 times per minute, a relaxation of the limits is allowed of 20 log 30/ <i>N</i> dB (where <i>N</i> is the number of clicks per minute). Criteria for separated clicks may be found in CISPR 14-1:2016.						
^b At transitional frequencies, the lower limit applies.						

Table 14 – Emission – Telecommunications/network port [Table 3 of IEC 61000-6-4:2011 (ed.2.1)]

Table clause	Port	Frequency range	Limits	Basic standard	Applicability note	Remarks
3.1	Telecommunications/ network	0,15 MHz to 0,5 MHz	97 dB(μV) to 87 dB(μV) quasi-peak 84 dB(μV) to 74 dB(μV) average 53 dB(μA) to 43 dB(μA) quasi-peak 40 dB(μA) to 30 dB(μA) average The limits decrease linearly with the logarithm of the frequency	CISPR 22:2008	See ^a and ^b	
		0,5 MHz to 30 MHz	87 dB(μV) quasi-peak 74 dB(μV) average 43 dB(μA) quasi-peak 30 dB(μA) average			
^a The current and voltage disturbance limits are derived for use with an impedance stabilization network (ISN) which presents a common mode (asymmetric mode) impedance of 150 Ω to the telecommunication port under test (conversion factor is 20 log ₁₀ 150 / I = 44 dB).						
^b When performing measurement using an ISN, the EUT shall meet the voltage limits of this table. All elements within CISPR 22:2008 shall be followed, including but not limited to selection of test method, test configuration, cable characteristics.						

NOTE For the transmission line side high frequency interface the recommended conducted emission limits in the frequency range used by APLC are given in Clause 5 and Figure 11.

12.2.2 Low frequency disturbance emission

If the terminal is powered by a DC power supply source, in order to verify the limits of LF conducted disturbance generated by the terminal which may affect other terminal connected to the same DC power source, the noise measured across the power supply terminals of the terminal under test shall not be greater than 3mV, psophometrically weighted, or 10 mV peak to peak.

The measurement must be carried out using a filter across the DC power source used as in Figure 24, where C1 is a ceramic or plastic film capacitor and C2 an electrolytic capacitor.

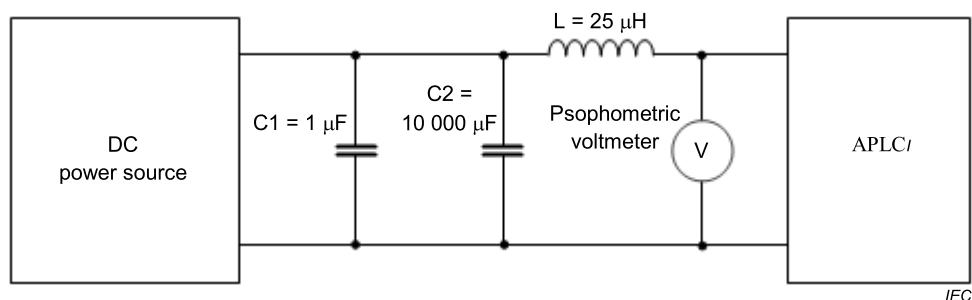


Figure 24 – LF disturbances measurement setup

12.3 Immunity

12.3.1 EMC Environment

Due to the fact that IEC 61000-6-5 adds some particular tests, and gives greater or equal levels of severity for the tests common with IEC 61000-6-2 it is requested for the immunity test APLC terminal shall comply with the requirements of IEC 61000-6-5.

Table 6 of IEC 61000-6-5:2015 shows the electromagnetic phenomena.

Typically APLC terminals are installed in the control room area according to IEC 61000-6-5.

Table 15 – Characterization of the electromagnetic phenomena
[Table 1 of IEC 61000-6-5:2015]

Continuous phenomena	Transient phenomena with high occurrence	Transient phenomena with low occurrence
Voltage variations: – a.c. power supply – d.c. power supply ^a Harmonics, interharmonics ^a Signalling voltages ^a Ripple on d.c. power supply Power frequency variation ^a Conducted disturbances in the range 2 kHz to 150 kHz ^a Conducted disturbances in the range 1,6 MHz to 30 MHz ^a Power frequency magnetic field (according to IEC 61000-4-8) Radiated, radio frequency electromagnetic field Conducted disturbances, induced by radio-frequency fields Mains frequency voltage (according to IEC 61000-4-16)	Voltage dips (duration ≤ 0,02 s): – a.c. power supply – d.c. power supply Voltage fluctuations Fast transient/burst Damped oscillatory/ring wave Damped oscillatory magnetic field Electrostatic discharge	Voltage dips (duration > 0,02 s): – a.c power supply – d.c. power supply Voltage interruptions: – a.c. power supply – d.c. power supply Short duration power frequency variation ^{a, b} Surge Short duration power frequency voltage Short duration power frequency magnetic fields (according to IEC 61000-4-8) Radiated pulsed disturbances
^a Not covered in this standard by dedicated immunity requirements. ^b In case of islanded systems (e.g. not connected to a public network), the characterization of the phenomenon changes from "low occurrence" to "high occurrence".		

For easy application of this document it is essential to classify the APLC signal ports according to Figure 3 and Figure 4 and the definitions given in IEC 61000-6-5:2015.

Table 16 – Port classification

DPLC terminal interface	Type of signal ports according to IEC 61000-6-5:2015
HF Transmission Line	4
Audio frequency / Voice	1
Subscriber	2
Data – within the control room	2
Data – other	3
LAN Connection	2
Binary I/O integrated teleprotection	4
External teleprotection	2
Alarm relays	2
Time information	1
Configuration port	1

12.3.2 Functional requirements

The tests shall be carried out with the terminal in operation. The terminal shall not suffer permanent damage from the tests, and after the tests the terminal shall be checked for correct operation.

The performance criteria for APLC are defined on the basis of the functions "Teleprotection" and "Data transmission and telecommunication" of IEC 61000-6-5:2015 (Table 17):

Table 17 – Performance criteria

Performance criteria	Description
A	The equipment shall continue to operate as intended during and after the test. No degradation of performance or loss of function is allowed
B	Temporary loss of function or degradation of performance which ceases after the disturbance ceases, and from which the equipment under test recovers its normal performance, without operator intervention

The APLC terminal functions to be considered for EMI testing are at least the followings:

- Protection and teleprotection
- Data transmission and telecommunication

12.3.3 Immunity test list

The following Table 18 summarizes both the immunity test to be performed as well as the corresponding levels to be used according to IEC 61000-6-5 and IEC 61000-6-2 (Surge test levels for "Power supply DC" and "Telecommunication").

Table 18 – Immunity test list

Test	Standard	Immunity Level						Performance Criteria
		Enclosure	AC Power Supply	DC Power Supply	Signal / Control Ports			
					1	2	3 & 4	
Electrostatic discharges	IEC 61000-4-2	6 kV (Contact) 8 kV (Air)						B
Radiated Electromagnetic field	IEC 61000-4-3	10 V/m 80 MHz – 1 GHz 1kHz, 80 % AM						A
		3 V/m 1 GHz – 2.7 GHz 1kHz, 80 % AM						A
		1 V/m 2.7 GHz – 6 GHz 1kHz, 80 % AM						A
Fast transient/burst	IEC 61000-4-4		2 kV ^(a) 5 kHz	2 kV ^(a) 5 kHz	1 kV ^(b) 5 kHz	2 kV ^(b) 5 kHz	4 kV ^(b) 5 kHz	A, B according to notes ^(a) and ^(b) respectively
Surge	IEC 61000-4-5		2 kV in CM ^(a) 1 kV in DM ^(a)	2 kV in CM ^(a) 1 kV in DM ^(a)	1 kV in CM ^(b) 0.5 kV in DM ^(b)	2 kV in CM ^(b) 1 kV in DM ^(b)	4 kV in CM ^(b,c,e) 2 kV in DM ^(b,c,e)	A, B according to notes ^(a) and ^(b) respectively
Conducted disturbances, induced by radio frequency fields	IEC 61000-4-6		10 V 150kHz – 80MHz 1kHz, 80 % AM	10 V 150kHz – 80MHz 1kHz, 80 % AM	3 V 150kHz – 80MHz 1kHz, 80 % AM	10 V 150 kHz – 80MHz 1kHz, 80 % AM	10 V 150kHz – 80MHz 1kHz, 80 % AM	A
Power Frequency Magnetic fields	IEC 61000-4-8	100 A/m cont. 1000 A/m 1 sec.						A
AC voltage dips and interruptions	IEC 61000-4-11		70 %UT, 1 period					A
			0 % UT, 5 periods					B
Mains frequency voltage (50/60 Hz)	IEC 61000-4-16			30 V Continuous ^(d)		30 V Continuous ^(d)	30 V Continuous ^(d)	A
				300 V for 1 s ^(d)		300 V for 1 s ^(d)	300 V for 1 s ^(d)	B

Test	Standard	Immunity Level						Performance Criteria
		Enclosure	AC Power Supply	DC Power Supply	Signal / Control Ports			
					1	2	3 & 4	
Ripple on DC power supply	IEC 61000-4-17			10 % U _N				A
Damped oscillatory wave	IEC 61000-4-18		2.5 kV CM ^(a) 1 kV DM ^(a) 1 MHz	2.5 kV CM ^(a) 1 kV DM ^(a) 1 MHz	1 kV CM ^(b) 0.5 kV DM ^(b) 1 MHz	1 kV CM ^(b) 0.5 kV DM ^(b) 1 MHz	2.5 kV CM ^(b) 1 kV DM ^(b) 1 MHz	A, B according to notes ^(a) and ^(b) respectively
DC voltage dips and interruptions	IEC 61000-4-29			70 % U _T , 100 ms 0 % U _T , 10 ms				A
				40 % U _T , 100 ms				B
CM = Common Mode (Line to Ground) DM = Differential Mode (Line to Line)								
(a) Performance criteria A								
(b) Performance criteria B								
(c) The surge waveform 10/700 μs is recommended for testing signal ports intended to be connected to telecom network or remote equipment via unshielded outdoor symmetrical communication lines. This surge form requires a different surge pulse generator and a different coupling impedance (e. g.40 Ω)..								
(d) Only the country specific mains frequency shall be used.								
(e) The surge waveform 10/700 μs is recommended for testing signal ports intended to be connected to telecom network or remote equipment via unshielded outdoor symmetrical communication lines.								

NOTE During immunity tests of telecommunication ports according to IEC 61000-4-6 the performance criterion A is only applicable if test signals contain frequency components outside the nominal receive and transmit high frequency band (B_{nom}) of DUT and outside of frequency guard intervals with the bandwidth $2B = 2B_{nom}$ on both sides of B_{nom} respectively. Otherwise the performance criterion B shall be used.

Annex A (normative)

Characteristics of compandors for telephony (based on the withdrawn ITU-T Recommendation G.162)

A.1 General

Compandors may be used in APLC systems to improve the quality of voice channels. The compandor consists of a compressor at the transmit side and an expander at the receive side. By compressing the voice signal at the transmit side a better signal-to-noise ratio can be achieved.

A.2 Characteristics of compandors

The following definitions are taken from the withdrawn ITU-T Recommendation G.162 to explain the characteristics of compandors.

A.3 Definition and value of the unaffected level

The unaffected level is the absolute level, at the reference point on the line between the compressor and the expander of a signal at 800 Hz, which remains unchanged whether the circuit is operated with the compressor or not. The unaffected level is defined in this way in order not to impose any particular values of relative level at the input to the compressor or the output of the expander.

The unaffected level shall be 0 dBm0.

A.4 Ratio of compression and expansion

- **Definition and preferred value of the ratio of compression**

The ratio compression of a compressor is defined by the formula:

$$\alpha = \frac{n_e - n_{e0}}{n_s - n_{s0}}, \quad (\text{A.1})$$

where

n_e Is the input level,

n_{e0} Is the input level corresponding to 0 dBm0;

n_s Is the output level;

n_{s0} Is the output level corresponding to an input level of n_{e0} .

The preferred value of α is 2.

- **Definition and preferred value of the ratio of expansion**

The ratio expansion of an expander is defined by the formula:

$$\beta = \frac{n'_s - n'_{s0}}{n'_e - n'_{e0}}, \quad (\text{A.2})$$

where

n'_e is the input level,

n'_{e0} is the input level corresponding to 0 dBm0;

n'_s is the output level;

n'_{s0} is the output level corresponding to an input level of n'_{e0} .

The preferred value of β is 2.

A.5 Range of level

The range of level over which the recommended value of α and β apply, shall extend at least:

- from +5 to –45 dBm0 at the input of the compressor, and
- from +5 to –50 dBm0 at the nominal output of the expander.

A.6 Signal to noise ratio

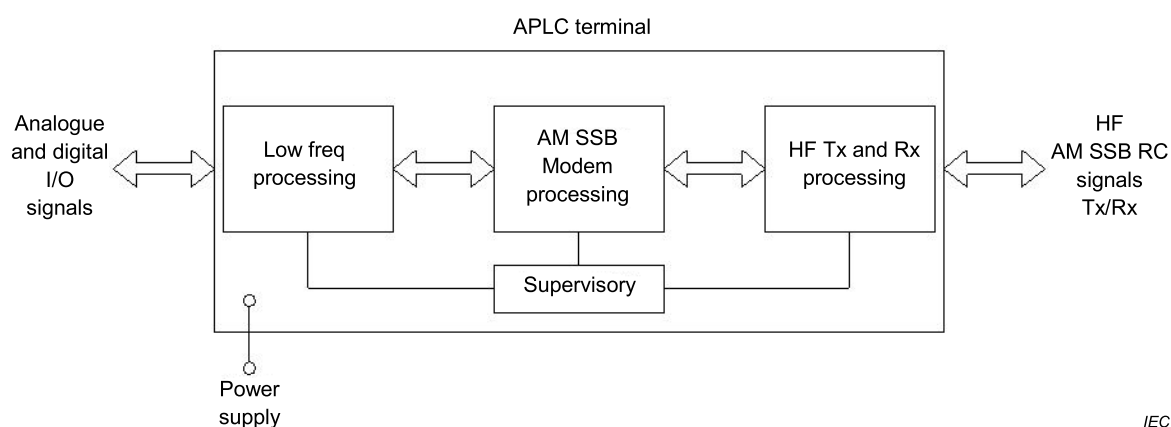
With the use of compandors, the improvement of signal-to-noise ratio will be approximately 15 dB.

Annex B (informative)

APLC communication model

B.1 General

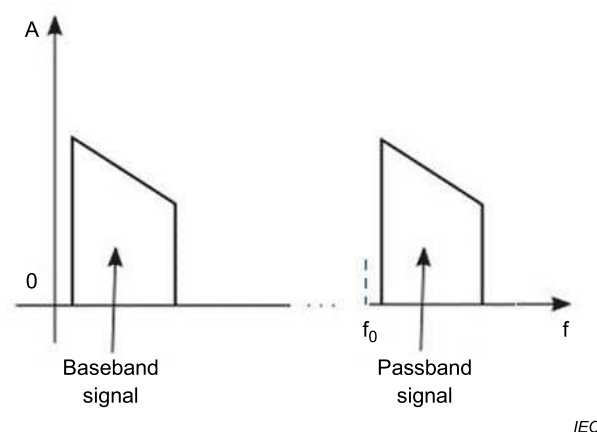
It is valuable to define the key features of the APLC terminal. This classification refers to the analogue modulation used to form the modulated signal which is for this specific Part 2 of this family standard exclusively AM-SSB as shown in Figure B.1.



IEC

Figure B.1 – Basic components of the APLC Terminal

This modulation falls in the class of linear modulations because the shape of the modulating signals is maintained and not affected by the modulation process. For SSB, the baseband signal and the modulated (pass band) signal, since it will be band limited around the carrier frequency f_0 , have the same bandwidth. For this reason it is the most bandwidth efficient way to transmit information across a channel. A comparison of the equivalent baseband version of a signal and its SSB-modulated pass band version around a carrier frequency f_0 is shown in Figure B.2.



IEC

**Figure B.2 – Baseband and pass band signals
correspondence in SSB modulation**

Figure B.3 shows the interfaces for low frequency, baseband and high frequency (pass band) signals in an APLC terminal.

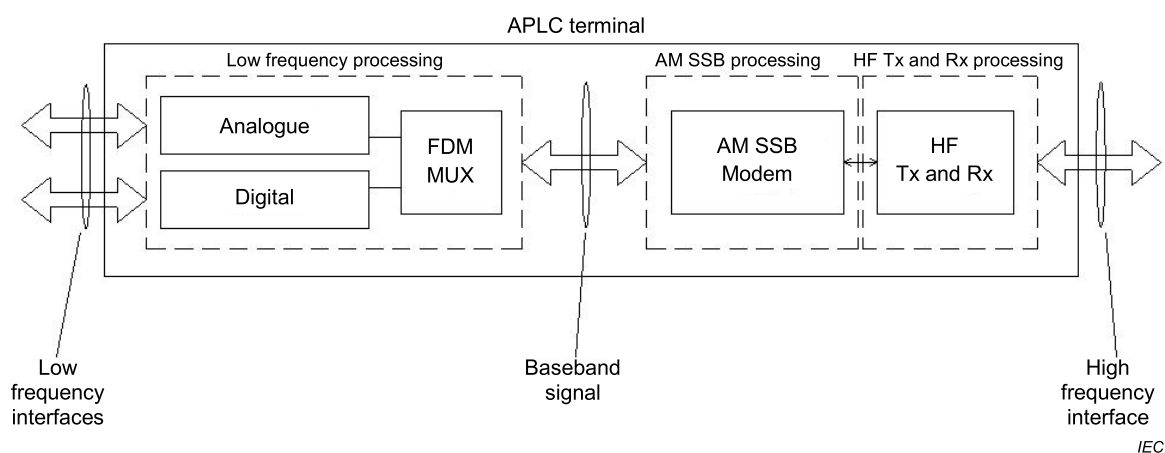


Figure B.3 – APLC Terminal LF, baseband and HF interfaces identification

This type of analogue modulation, along with frequency division multiplexing technique (FDM), has been successfully used over power lines for long time in order to effectively allocate each basic high frequency channel, allowing the implementation of both contiguous as well as separated band schemes.

The modulating baseband signal results from the aggregation of one low frequency signal, or more in case FDM is used. Examples are shown in Figures B.4 and B.5.

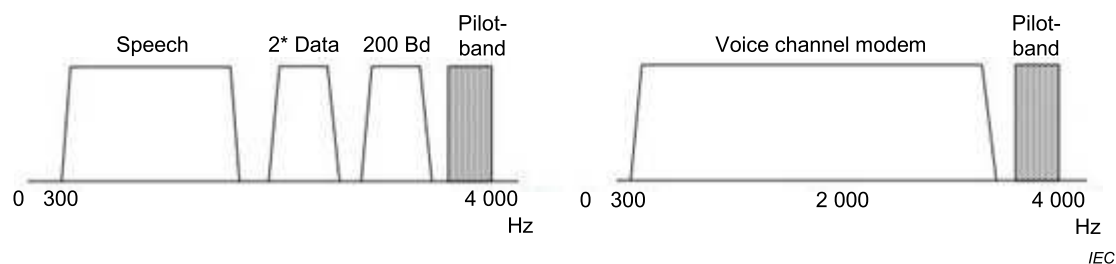


Figure B.4 – Examples for low frequency signals with bandwidth 4 kHz (IEC 62488-1)

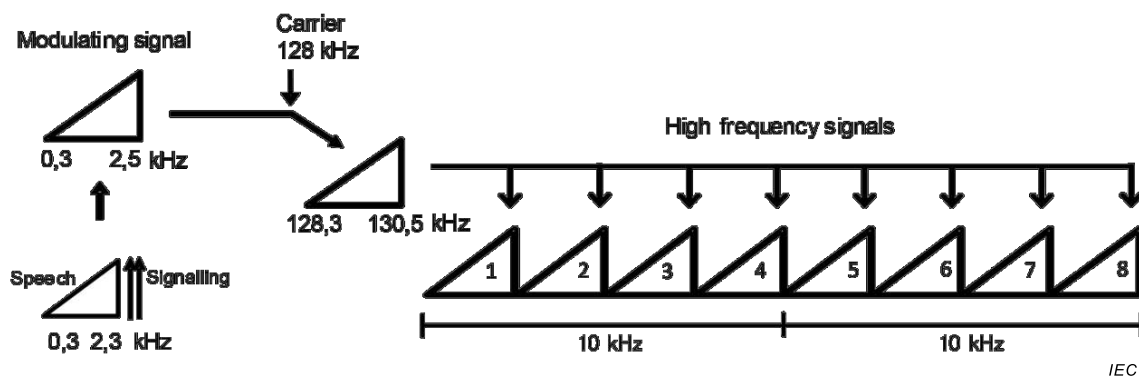


Figure B.5 – Composition of the modulating baseband for eight telephony channels with signalling APLC terminal (source Japan NC)

Indeed for this modulation, the frequency band allocated for the low frequency modulating signal, e.g. 4 kHz for the ITU-T basic single voice channel, or 20 kHz for the eight 2,5 kHz

voice channels as shown in Figure B.5, is exactly the same after the modulation process but the spectrum is shifted in the upper or lower sideband of the carrier frequency.

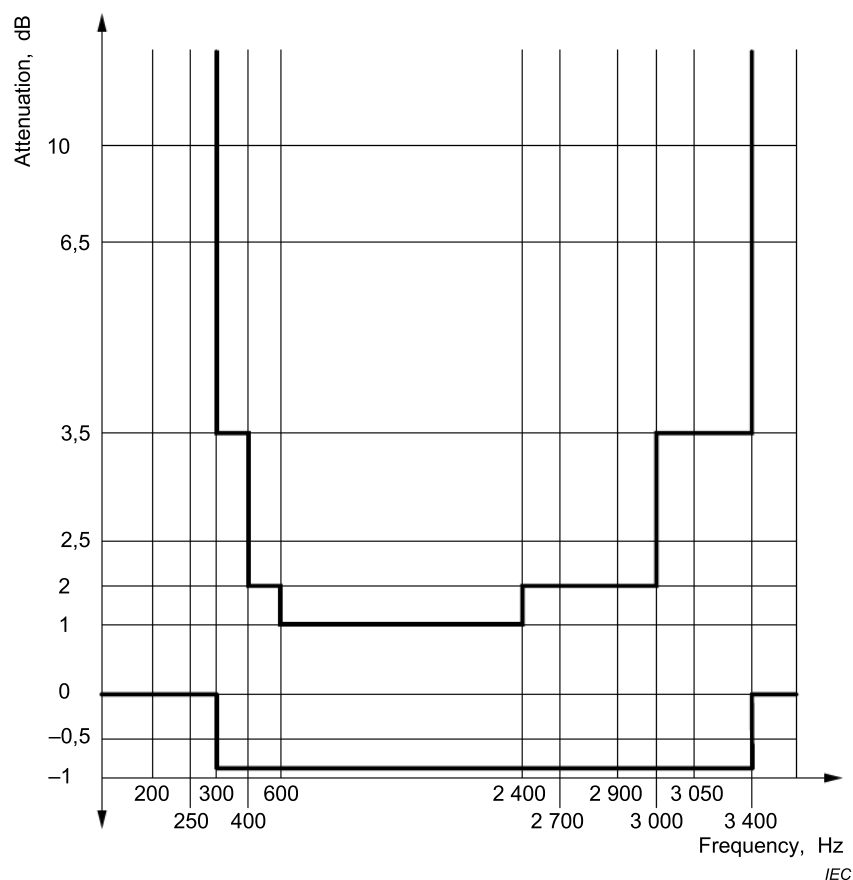


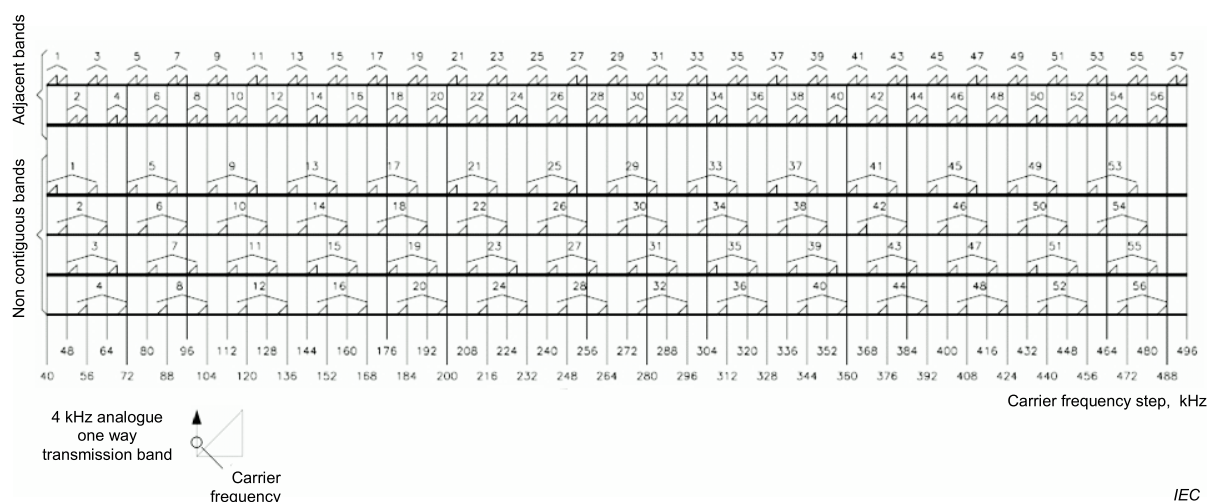
Figure B.6 – Line-up limits of circuits for a 4 kHz channel terminal (ITU-T G.120)

It is important to define the correct levels of each component of the aggregate baseband signal to avoid overload and distortion.

Figure B.6 shows for example the line-up mask defining the limits of circuits with 4 kHz channel terminal according to ITU-T G.120.

Thanks to the subsequent frequency translation process the AM-SSB envelope modulation is then frequency-shifted up (up-conversion at the transmitter or down-conversion at the receiver) to the right HF channel.

Figure B.7 shows an example of a high frequency channeling plan, based on a 4 kHz basic frequency band allowing both the non-contiguous as well as the adjacent allocation for the transmission and reception bands.



**Figure B.7 – Example of HF channelling plan
(4 kHz based – IEC 62488-1)**

B.2 AM-SSB modulation technique

In the technical literature there are a number of well-known methods to implement AM-SSB modulation both traditional as well as digital processing based.

A detailed description is out of the scope of this document, but in order to understand the main functional characteristics, some basic information is presented.

For instance, one possible method for SSB generation is the phasing method which uses two balanced modulators and phase shifting that causes one of the side bands to be cancelled out.

Figure B.8 shows the simplified structure of a possible AM-SSB modulator used in APLC terminals.

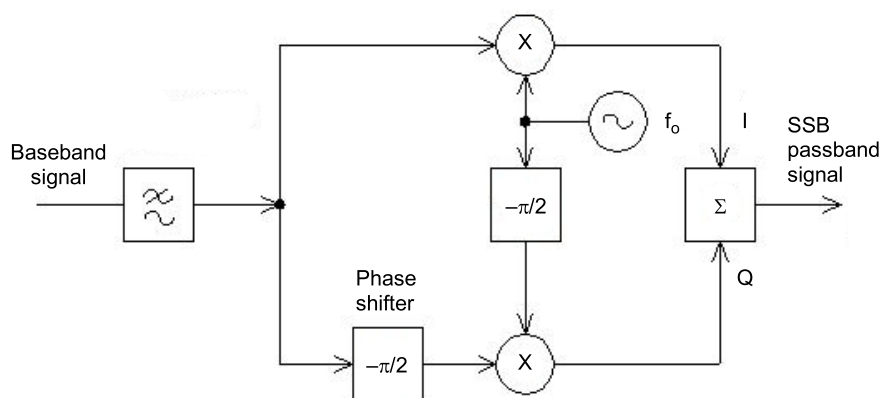


Figure B.8 – Principle of phasing SSB modulator

Though this modulation uses substantially less bandwidth and power, it cannot be demodulated by a simple envelope detector in the same way as in the standard amplitude modulation.

Figure B.9 gives an example of phasing SSB demodulator.

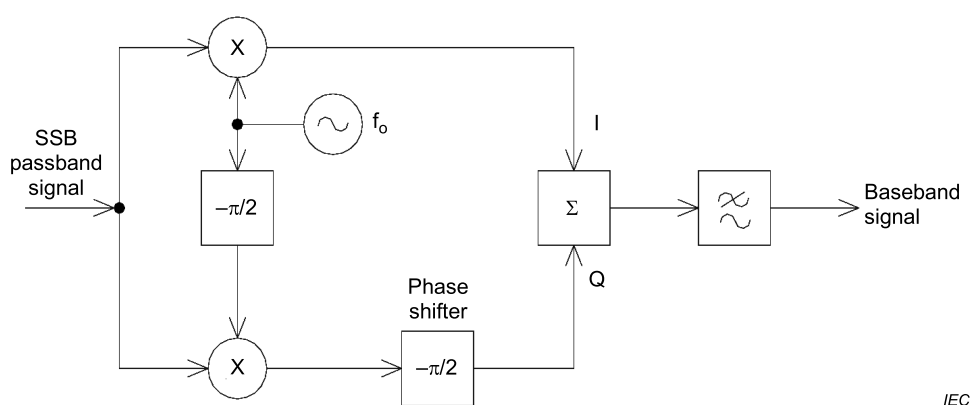


Figure B.9 – Principle of phasing SSB demodulator

The process for recovering the correct, in principle not transmitted carrier frequency f_0 , introduces some complexity in the receivers design.

One approach to solving this problem is to add a small sinusoidal component called a pilot tone whose frequency has a known relationship to the carrier frequency.

The pilot tone frequency might be chosen to be the carrier frequency when the baseband signal has no DC components.

The receiver will then generate a local carrier reference by using a narrow bandwidth band-pass filter to select the pilot tone and use it to restore correctly the frequency of the locally generated carrier.

The following features of AM-SSB are relevant:

- This modulation allows better management of the frequency spectrum. More information can fit into a given frequency band than would be possible with other amplitude modulation techniques;
- Most of the transmitted power is concentrated on one single sideband and little is spent as carrier power and no power for the redundant sideband;
- It uses the minimum bandwidth and therefore also minimizes the noise within the signal bandwidth.

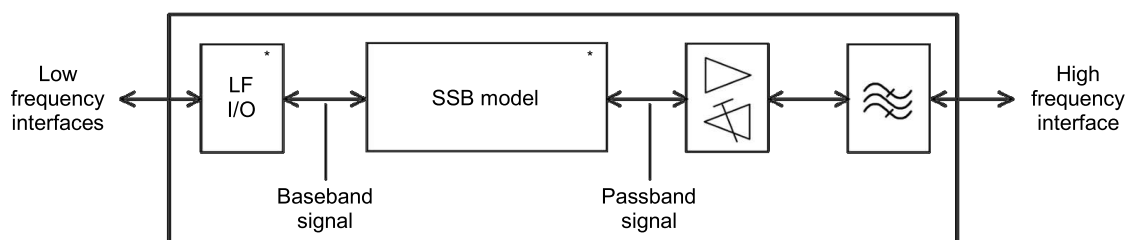
However, care must be taken, especially for the multichannel APLC terminals, not to produce harmful products caused by non-linearity of its amplifiers.

B.3 Functional blocks of an APLC terminal

The most important part in designing APLC transmission systems is the processing of the signals.

All the processes shown in the communication model of these systems described in the past were done using exclusively analogue techniques. Because of this, the volume and the production process were complex and costly.

Figure B.10 shows the APLC functional blocks typically implemented using DSP techniques.



IEC

*) Nowadays these functional blocks are implemented using DSP techniques.

Figure B.10 – Generic APLC terminal main functional blocks

Thanks to the introduction of digital signal processing (DSP), even if the output of APLC system remains analogue, all the processes are internally performed digitally in which case the precision of signal processing is high, giving all the advantages of digitized systems, i.e. direct HF channel allocation, advanced supervisory control, temperature stability.

As a result, this structure brings low cost, high integration, high precision, high reliability and is easier to manage.

Annex C (informative)

HF modulated power signal

C.1 General

In an APLC system, various signals such as voice, data transmission using discrete tone, teleprotection and pilot signals are modulated simultaneously, occupying a different range of HF frequencies. The total RMS power can be expressed by the following equation;

$$P_{RMS} = \sum_i^{n_1} p_i(f_i) + \int_{B_n} p_{RMS}(f) df + \sum_j^{n_2} q_j(w_j) \quad (C.1)$$

where, $p_i(f_i)$ are power of discrete tone signals at frequencies f_i , $P_{RMS}(f)$ is the continuous part of the spectrum in the nominal power frequency band, and $q_j(w_j)$ represents the power of individual voice channels, the total power of which is stochastically evaluated, B_n is the used PLC band, where the power assigned to the band is assumed to be proportional to its bandwidth, n_1 is the number of discrete tone signals and n_2 is the number of voice channels.

The first term and the second term of equation (C.1) are calculated in a deterministic way, but the third term requires a complicated stochastic summation of the power of individual channels.

In actual systems, only one term, or a combination of any two terms of equation (C.1) may be present. Also a variety of input signals with various levels and bandwidths form a wide range of input signal interface line-up. The following information only gives fundamental aspects of tone and voice signals to find out appropriate ways to determine the load capacity of APLC systems, and to give a way to distribute output power to each signal.

Since various input signals are loaded simultaneously into the APLC terminal, incidental combination of high voltages may cause an instantaneous high peak power that produces harmful spurious emissions and/or inter-channel interference due to the non-linearity of the HF amplifier. In this respect, the range of linearity to be secured and also the limit of the non-linearity to be permitted become very important factors for designing an amplifier.

If the number of input signals is small, the sum-of-voltage rule can be applied as a safe side procedure. On the other hand, if its number is large, it requires excessive power for the amplifier to follow the sum-of-voltage rule, and therefore, the sum-of-the-power rule or the method of stochastic sum-of-the-power should be applied instead. The total RMS power of the combined signals can be calculated easily by equation (C.1), but it may entail difficult stochastic approaches to determine the total load capacity and the PEP of the system.

C.2 Discrete tone signals

When a channel is loaded with a number of sine waves, it is important to evaluate the probability of all the amplitudes added in phase. When the number of sine waves is small, this in-phase addition probability is large, but it decreases rapidly when the number exceeds several waves.

The probability of exceeding a certain level of peak power produced by the combined waves can be calculated by the convolution integral of the following sine wave probability-density-function

$$f(x) = \frac{1}{\pi\sqrt{a^2 - x^2}}, \quad -a \leq x \leq a \quad (\text{C.2})$$

where x is the sine wave voltage and a is the amplitude of the sine wave.

The mean power of a sine wave is equal to its variance σ^2 , and the mean power of the n combined sine waves becomes $n\sigma^2$.

Figure C.1 shows the sine wave and its probability distribution.

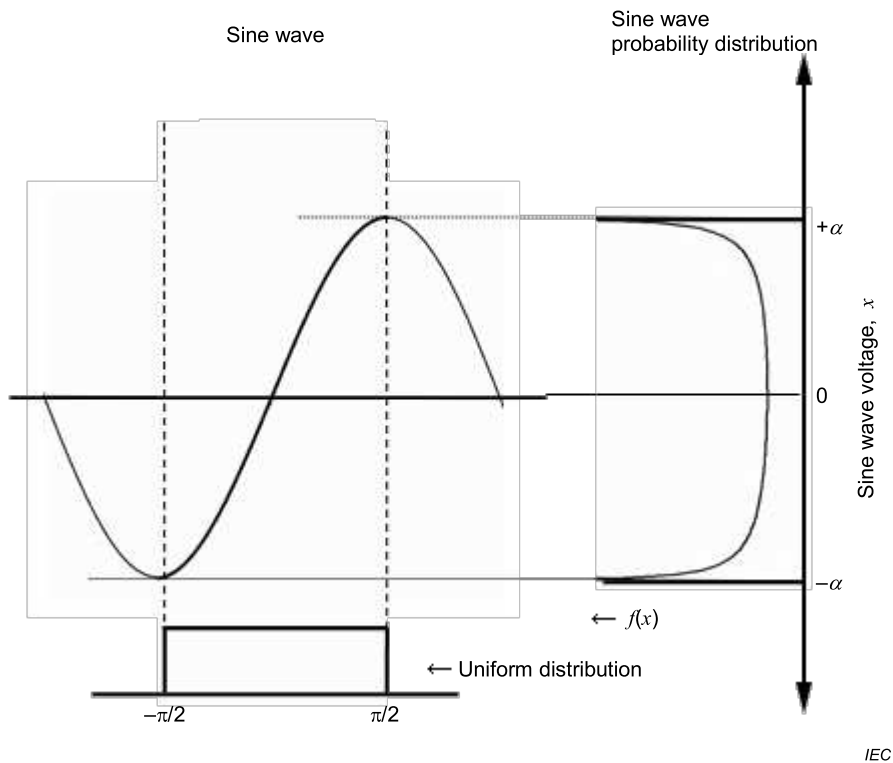


Figure C.1 – Sine wave and its probability distribution

The convolution of equation (C.2) can be applied irrespective of whether the same frequencies or different frequencies are used.

Figure C.2 shows the exceedance probability of 0,1 %, and 1 % of the combined distribution together with $n^2\sigma^2$, $n\sigma^2$, $4n\sigma^2$, and also the curve defined in ITU-R SM.326-7 (see the note below).

The curve of ITU-R SM.326-7 is close to the 1 % curve of the combined distribution; and the level of which seems considerably safe.

NOTE In order to obtain the mean power of a telegraph channel, following two equations are given in ITU-R SM.326-7, where n is the number of telegraph sub-channels of identical level.

$$P_{\text{mean}} = \frac{P_{\text{peak}}}{n^2}, \quad \text{when } n = 1, 2, 3 \text{ or } 4,$$

$$P_{\text{mean}} = \frac{P_{\text{peak}}}{4n}, \quad \text{when } n > 4,$$

where

P_{mean} = Mean power

P_{peak} = Peak envelope power allocated to the group of channels

n = number of telegraph sub-channels of identical level

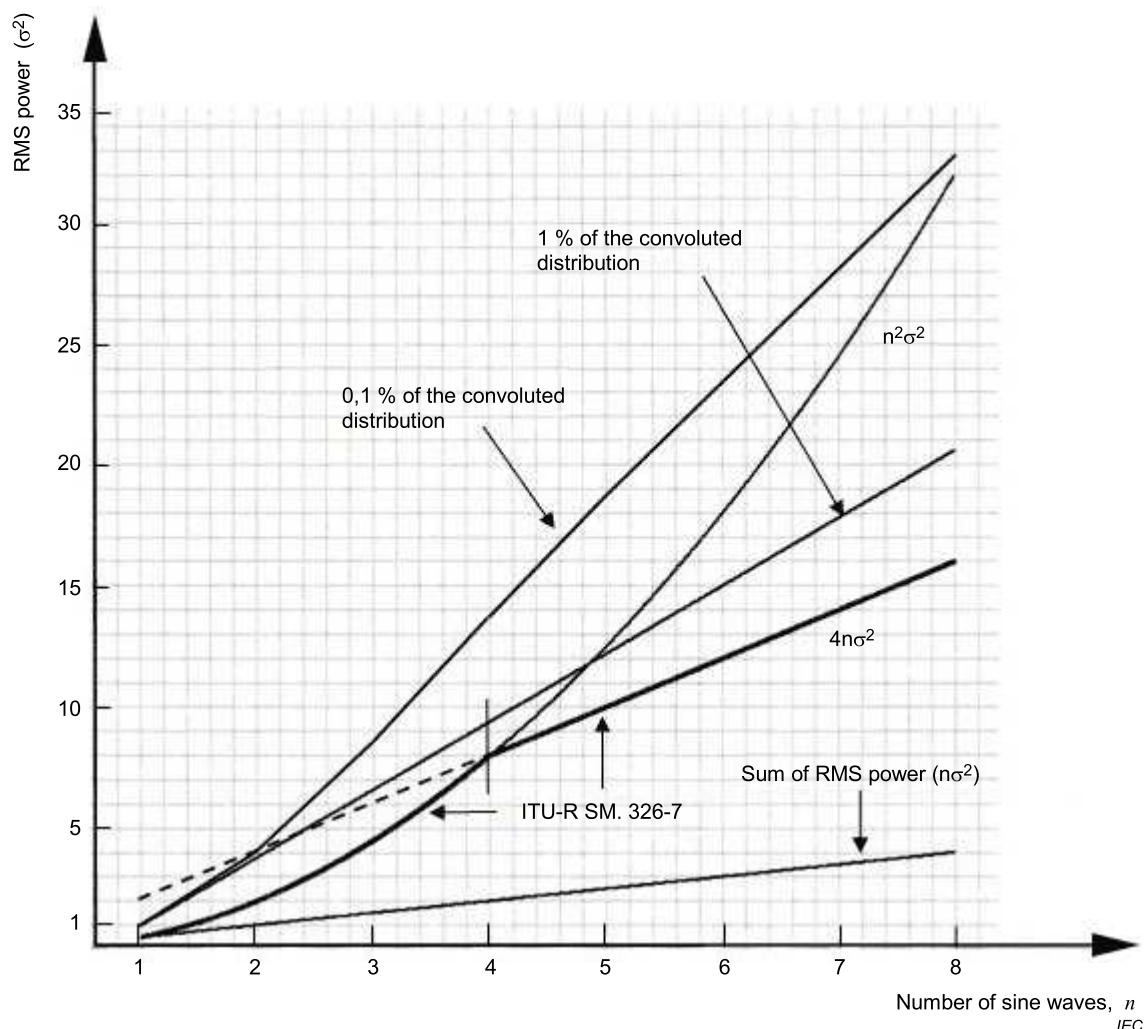


Figure C.2 – Probability of combined sine waves

C.3 Voice channels

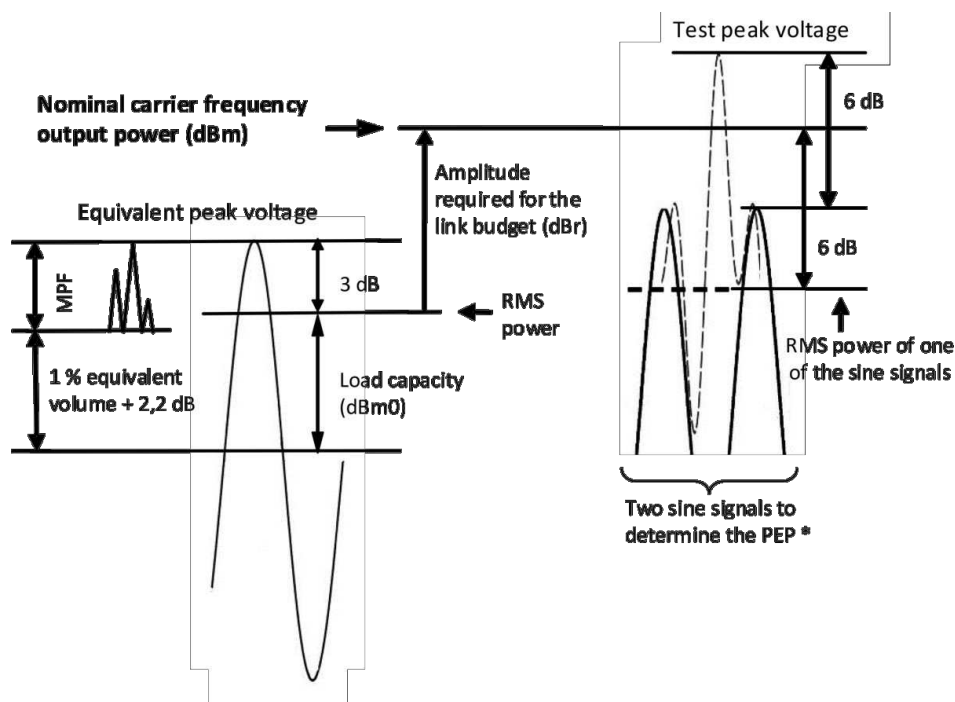
For calculating the load capacity of the multichannel terminals with four or more channels, the method of the stochastic sum-of-the-power must be applied according to the concept of ITU-T G. 223.

In ITU-T G.223 (11/88), subclause 6.2, the equivalent RMS sine signal power whose amplitude is identical to the peak voltage of the multiplex signal is defined. This power is derived from the theory by Holbrook and Dixon [1]¹, although the mean power per channel is increased to -15 dBm0 from the original -16 dBm0.

The nominal high frequency band output power of multichannel PLC terminals can be determined from this power by adding necessary amplitude required for the link budget; and measured by applying sine signals according to the test procedures described in 7.6 and 7.8.

¹ Numbers in square brackets refer to the bibliography.

The related structure is schematically illustrated in Figure C.3.



IEC

MPF: Multi-channel peak factor
Limiting ration of the overload voltage to the RMS voltage for a given number of active channels

Equivalent volume: Volume related to the average multi-channel speech power

Load capacity: Refer to 2.2.14

* In the case of two tone testing method

Figure C.3 – Nominal high frequency band output power of multichannel PLC terminals

C.4 Composite channels

When a terminal is modulated with some discrete frequencies and one or two voice channels, calculation of the first term and second term of the equation (C.1) can be used, and thus, the procedure stated in Subclause 6.2 of IEC 62488-1:2012 can be used. This procedure is advantageous because no stochastic approach is involved. However, for a large number of voice channels, i.e. four voice channels or more, the use of the third term of the equation (C.1) becomes inevitable.

One of the methods to determine the peak power of multi-channel signals modulated partially with discrete tone signals is to let the total RMS power of the tone signals be less than that of the average power of the single voice channel, and calculate the total peak power under the assumption that all channels are composed of voice channels. This method is used when the number of discrete tone signals is small. For example, according to the paper by Holbrook and Dixon [1], the peak to average ratio of a single voice channel is 18,5 dB whereas the 0,1 % peak to average ratio of eight sine waves convoluted together is 9,3 dB (see the note below). The difference between them is enough to neglect the influence by discrete tones. (This method is being used in Japan.)

However in this method, the level of each tone signal has to be decreased in accordance with the increase of the number of tone signals, resulting in a degraded signal to noise performance.

NOTE From Figure C.2, the 0,1 % peak to average ratio for 8 tone signals equals $10 \times \log_{10}(33,7/4) = 9,3$ dB.

For higher levels of tone signals, a realistic way to solve this problem is shown in the literature [3]. In order to derive the total load capacity, this method is to apply the MPF (multichannel peak factor) of the voice channels to the power sum of total RMS voice load and the total RMS tone signal load.

The load capacity of the voice channels is represented by equation (C.3):

$$L_C = 10 \log_{10} N + L_{V1} + \Delta C_1 + \Delta C_2 - 3 + M_a \text{ (dBm0)} \quad (\text{C.3})$$

where

N : Number of voice channels in the system,

L_{V1} : Mean power of one voice channel, assumed to be –15 dBm0 according to ITU-T G.223,

ΔC_1 : Increment factor from mean power to 99 % value of the voice power distribution,

The value of ΔC_1 can be derived from the literature [1],

$$\Delta C_2 : \text{MPF (Multichannel peak factor), } MPF_n = 12,9 + \frac{6}{1 + 0,07n} \text{ (dB),}$$

(Quoted from ITU-T G.Sup22),

n : Number of active channels,

M_a : Safety margin

The relation between the number of system channels N and the number of estimated active channels n is given as $N\tau$ taking round-down integer, where τ is the largest probability of the busiest hour that a channel may be active.

$$-3 \text{ dB: Peak to average ratio of the sine wave in dB} = 20 \log_{10}(1/\sqrt{2}).$$

Table C.1 shows the computation process of the load capacity for the N channels system composed of only voice channels.

Table C.1 – Load capacity of voice channels

Number of system channels N (active channels n)	4 (1)	6 (1)	8 (2)
$L_{V2} = 10 \log_{10} N + L_{V1}$ (dBm0), $L_{V1} = 15$ (dBm0)	–9,0	–7,2	–6,0
ΔC_1 (dB)	10,0	9,0	8,8
ΔC_2 (dB)	18,5	18,5	18,2
$L_C = L_{V2} + \Delta C_1 + \Delta C_2 - 3$ (dBm0)	16,5	17,3	18,0

The method to determine the total load capacity L_T suggested by Bell Laboratories [2] is to add the total RMS power of tone signals $P_{RMS(n_1)}$ to the mean power of N voice channels L_{V2} , and then apply the $\Delta C_1 + \Delta C_2 - 3$ shown as follows:

$$L_T = L_m + \Delta C_1 + \Delta C_2 - 3 \text{ (dBm0)} \quad (\text{C.4})$$

where

$$L_m = 10 \log_{10} \left(10^{L_{V2}/10} + 10^{P_{RMS(n_1)}/10} \right),$$

$$P_{RMS(n_1)} = \sum_{i=1}^{n_1} p_i(f_i),$$

n_1 = number of tone signals.

Equation (C.4) can be used to distribute the total RMS signal power to each signal by changing the values of L_{V2} and $P_{RMS(n_1)}$, but keeping the value of L_m unchanged (see C.5.2 Calculation Example 1).

C.5 Calculation examples

C.5.1 General

The safety margin M_a is not considered in the following examples.

C.5.2 Calculation example 1: Load capacity and PEP

Determine the total load capacity when the total RMS tone signals of -12 dBm0 is loaded to the $N = 4$ channel system.

- Total RMS mean power: $L_m = 10 \log_{10} (10^{-9/10} + 10^{-12/10}) = 10 \log_{10} (0,19) = -7,2 \text{ (dBm0)}$
- $L_T = L_m + \Delta C_1 + \Delta C_2 - 3 = -7,2 + 10 + 18,5 - 3 = 18,3 \text{ (dBm0)},$

The resultant L_T is larger than the load without the tone signals L_C by 1,8 dB

The PEP of this system can be calculated by adding the power required by the link budget calculation according to Subclause 6.2 of IEC 62488-1:2012 to this load capacity.

If the power required by the link budget is 27 dBr, then the PEP becomes $18,3 + 27 = 45,3 \text{ dBm}$.

C.5.3 Calculation example 2: Power distribution adjustment

- Number of voice channels in the system: $N = 4$
- PEP = 45,3 dBm
- Total RMS power of discrete tones = -10 dBm0 (higher than the above Example 1 by 2 dB)
- Total signal RMS power = $\text{PEP} - \Delta C_2 + 3 = 45,3 - 18,5 + 3 = 29,8 \text{ dBm}$

The level of 29,8 dBm0 can be distributed to the combined voice signals and combined discrete tone signals as follows:

- True value of total signal RMS power: $= 10^{\frac{29,8}{10}} = 955$

- True value of combined discrete tone signals: $= 10^{\frac{-10}{10}} = 0,1$
- Distributed discrete tone signals $a = 955 \times \frac{0,1}{0,1 + (0,19 - 0,1)} = 502,6 \rightarrow 27,0 \text{ dBm (52,6 \%)}$
- Distributed voice signals $= 955 \times \frac{0,19 - 0,1}{0,1 + (0,19 - 0,1)} = 452,4 \rightarrow 26,6 \text{ dBm (47,4 \%)}$

The value 0,19 is the total RMS mean power in Example 1.

This combined power of the discrete tones can be further divided into individual tone signals according to power sum principle.

The literature [2] says that the method stated above gives an approximate load capacity and may not always be correct for particular cases. It is, therefore, important to allow some margin, and for special cases, it may be required to conduct a relevant performance test.

For references specific to this annex, refer to the bibliography ([1], [2], [3]).

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